



Quantum Mechanics

量子力学

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1 形式理论

仅介绍不基于泛函分析的非严格有限维 Hilbert 空间形式理论, 默认算符的定义域为整个 Hilbert 空间.

1.1 五大公设

1.1.1 态公设

Post. 1 态矢量

- 一个量子系统的物理状态, 可由 Hilbert 空间中的一个态矢量 $|\psi\rangle \in \mathcal{H}$ 所完全描述.
- 物理状态对应的态矢量不唯一, 实际对应一个等价类 $|\psi\rangle \sim c|\psi\rangle, \forall c \in \mathbb{C}^*$.
- 通常选取归一化态矢量 $\langle\psi|\psi\rangle = 1$ 作为代表元, 而其整体相位不可观测

$$|\psi\rangle \sim e^{i\varphi} |\psi\rangle \quad (1.1)$$

1.1.2 可观测量公设

Post. 2 可观测量算符

每一个可观测量 A 均对应 Hilbert 空间上的一个自伴算符 $\hat{A} = \hat{A}^\dagger$.
从而, 记可观测量为 $\hat{A}$, 可观测量的观测值为 A .

1.1.3 测量公设

Post. 3 测量结果公设

测量可观测量 $\hat{A}$ 时, 测量结果 (measurement outcome) 一定是 $\hat{A}$ 的本征值. 有本征方程:

- 分立

- 非简并

$$\hat{A} |a_n\rangle = a_n |a_n\rangle \quad (1.2)$$

- 简并

$$\hat{A} |a_n^{(i)}\rangle = a_n |a_n^{(i)}\rangle \quad (i = 1, 2, \dots, g_n) \quad (1.3)$$

- 连续

- 非简并

$$\hat{A} |a\rangle = a |a\rangle \quad (1.4)$$

– 简并

$$\hat{A}|a, r\rangle = a|a, r\rangle \quad (r = 1, 2, \dots, g(a)) \quad (1.5)$$

其中 $|a_n\rangle$ 为属于本征值 a_n 的本征态, $|a\rangle$ 为属于本征值 a 的本征态.

Post.4 Born 概率公设

- 若系统处于 $|\psi\rangle$ 态, 则可分别展开为

$$\begin{aligned} |\psi\rangle &\stackrel{\text{non-deg.}}{=} \sum_n c_n |a_n\rangle & c_n &= \langle a_n | \psi \rangle \\ &\stackrel{\text{deg.}}{=} \sum_{n,i} c_n^{(i)} |a_n^{(i)}\rangle & c_n^{(i)} &= \langle a_n^{(i)} | \psi \rangle \end{aligned} \quad (1.6)$$

$$\begin{aligned} |\psi\rangle &\stackrel{\text{non-deg.}}{=} \int da c_a |a\rangle & c_a &= \langle a | \psi \rangle \\ &\stackrel{\text{deg.}}{=} \int da \sum_r c(a, r) |a, r\rangle & c(a, r) &= \langle a, r | \psi \rangle \end{aligned} \quad (1.7)$$

- 测得 a_n 的概率为

$$\begin{aligned} P(a_n) &\stackrel{\text{non-deg.}}{=} |c_n|^2 = |\langle a_n | \psi \rangle|^2 \\ &\stackrel{\text{deg.}}{=} \sum_{i=1}^{g_n} |c_n^{(i)}|^2 = |\langle a_n^{(i)} | \psi \rangle|^2 = \langle \psi | \hat{P}_n | \psi \rangle \end{aligned} \quad (1.8)$$

其中 $\hat{P}_n = \sum_{i=1}^{g_n} |a_n^{(i)}\rangle \langle a_n^{(i)}|$ 为投影算符.

Rmk 若态矢量 $|\psi\rangle$ 未归一化, 则概率写为

$$P(a_n) = \frac{\langle \psi | \hat{P}_n | \psi \rangle}{\langle \psi | \psi \rangle} \quad (1.9)$$

- 测得 a 的概率密度为

$$\begin{aligned} \rho(a) &\stackrel{\text{non-deg.}}{=} |c(a)|^2 = |\langle a | \psi \rangle|^2 \\ &\stackrel{\text{deg.}}{=} \langle \psi | \hat{P}(a) | \psi \rangle \stackrel{\text{unnorm.}}{=} \frac{\langle \psi | \hat{P}(a) | \psi \rangle}{\langle \psi | \psi \rangle} \end{aligned} \quad (1.10)$$

其中 $\hat{P}(a) = \sum_{r=1}^{g(a)} |a, r\rangle \langle a, r|$ 为投影密度算符.

Rmk 测量值落在 a 附近 Δ 区间内的概率为

$$P(a, \Delta) = \frac{\langle \psi | \hat{P}_\Delta | \psi \rangle}{\langle \psi | \psi \rangle} \quad \hat{P}_\Delta = \int_\Delta da \hat{P}(a) = \int_\Delta da \sum_{r=1}^{g(a)} |a, r\rangle \langle a, r| \quad (1.11)$$

Post.5 坍缩公设

若测量 A 得到本征值 a_n , 则系统瞬间坍缩到对应本征态.

$$|\psi\rangle \begin{cases} \xrightarrow[\text{non-deg.}]{a_n} |a_n\rangle \\ \xrightarrow[\text{deg.}]{a_n} \frac{|\psi_n\rangle}{\sqrt{\langle\psi_n|\psi_n\rangle}} = \frac{\hat{P}_n |\psi\rangle}{\sqrt{\langle\psi|\hat{P}_n|\psi\rangle}} \end{cases} \quad (1.12)$$

$$|\psi\rangle \begin{cases} \xrightarrow[\text{non-deg.}]{a} |a\rangle \\ \xrightarrow[\text{deg.}]{a} \frac{|\psi_\Delta\rangle}{\sqrt{\langle\psi_\Delta|\psi_\Delta\rangle}} = \frac{\hat{P}_\Delta |\psi\rangle}{\sqrt{\langle\psi|\hat{P}_\Delta|\psi\rangle}} \end{cases} \quad (1.13)$$

1.1.4 演化公设

1.1.5 全同粒子与对称化公设

1.2 Hilbert 空间

1.2.1 复线性空间

Def.1 复线性空间

设 V 为一个非空集合, 其元素称为矢量, $\mathbb{C}$ 为复数域. 若在 V 上定义加法与纯量乘法运算

$$V \times V \rightarrow V \quad (|v\rangle, |u\rangle) \mapsto |v\rangle + |u\rangle$$

$$\mathbb{C} \times V \rightarrow V \quad (c, |v\rangle) \mapsto c|v\rangle$$

且满足运算法则

1. 加法交换律: $\forall |v\rangle, |u\rangle \in V,$

$$|v\rangle + |u\rangle = |u\rangle + |v\rangle$$

2. 加法结合律: $\forall |v\rangle, |u\rangle, |w\rangle \in V,$

$$(|v\rangle + |u\rangle) + |w\rangle = |v\rangle + (|u\rangle + |w\rangle)$$

3. 存在零矢量: $\exists 0_V \in V, \text{ s.t. } \forall |v\rangle \in V,$

$$|v\rangle + 0_V = 0_V + |v\rangle = |v\rangle$$

4. 加法逆元: $\forall |v\rangle \in V, \exists (-|v\rangle) \in V, \text{ s.t.}$

$$|v\rangle + (-|v\rangle) = (-|v\rangle) + |v\rangle = 0_V$$

5. 纯量乘法的单位元: $\forall |v\rangle \in V,$

$$1_{\mathbb{C}} |v\rangle = |v\rangle$$

6. 纯量乘法结合律: $\forall a, b \in \mathbb{C}, |v\rangle \in V,$

$$(ab) |v\rangle = a(b |v\rangle)$$

7. 纯量加法对纯量乘法的分配律: $\forall a, b \in \mathbb{C}, |v\rangle \in V,$

$$(a + b) |v\rangle = a |v\rangle + b |v\rangle$$

8. 纯量乘法对矢量加法的分配律: $\forall c \in \mathbb{C}, |v\rangle, |u\rangle \in V,$

$$c(|v\rangle + |u\rangle) = c |v\rangle + c |u\rangle$$

则称 V 为复数域 $\mathbb{C}$ 上的一个线性空间, 简称复线性空间 (complex linear space).

Prop. 1 复线性空间的性质

1. V 的零元 0_V 唯一.
2. $\forall |v\rangle \in V, |v\rangle$ 的负元唯一.
3. $\forall |v\rangle \in V, 0_{\mathbb{C}} |v\rangle = 0_V$.
4. $\forall c \in \mathbb{C}, c 0_V = 0_V$.
5. $\forall c \in \mathbb{C}, |v\rangle \in V$, 若 $c |v\rangle = 0$, 则 $c = 0_{\mathbb{C}}$ 或 $|v\rangle = 0_V$.
6. $\forall |v\rangle \in V, (-1_{\mathbb{C}}) |v\rangle = -|v\rangle$.

1.2.2 复内积空间

Def. 2 复线性空间上的内积

设 V 为一个复线性空间. 在 V 上定义二元函数

$$\langle | \rangle : V \times V \rightarrow \mathbb{C} \quad (|v\rangle, |u\rangle) \mapsto \langle v|u\rangle$$

若 $\forall |v\rangle, |u\rangle, |w\rangle \in V, a, b \in \mathbb{C}$, 满足

1. 共轭对称性

$$\langle v|u\rangle = \langle u|v\rangle^*$$

2. 对第二个矢量保持线性

$$\langle v|au + bw\rangle = a\langle v|u\rangle + b\langle v|w\rangle$$

3. 正定性

$$\langle v|v\rangle \geq 0$$

且 $\langle v|v\rangle = 0$ iff $|v\rangle = 0_V$.

则称 $\langle | \rangle$ 为 V 上的一个内积 (inner product).

Prop. 2

内积对第一个矢量保持共轭线性:

$$\langle av + bu|w\rangle = a^* \langle v|w\rangle + b^* \langle u|w\rangle$$

Def. 3 复内积空间

若复线性空间 V 指定了一个内积 $\langle | \rangle$, 则称 $(V, \langle | \rangle)$ 为一个复内积空间 (complex inner product space).

Prop. 3 矢量长度

定义 $|v\rangle \in V$ 的长度为 $\| |v\rangle \| = \sqrt{\langle v|v\rangle}$. $\forall c \in \mathbb{C}$, 满足

$$\| c |v\rangle \| = |c| \| |v\rangle \|^$$

Lem. 1 Cauchy-Schwarz 不等式

$\forall |v\rangle, |u\rangle \in V$,

$$|\langle v|u\rangle| \leq \| |v\rangle \| \| |u\rangle \|$$

当且仅当 $|v\rangle$ 与 $|u\rangle$ 线性相关时取等, $k \in \mathbb{C}$.

1.2.3 赋范线性空间

Def. 4 范数

设 V 为一个复线性空间. 定义正定函数

$$\| \cdot \| : V \rightarrow [0, \infty) \subset \mathbb{R}$$

若 $\forall |v\rangle, |u\rangle \in V, c \in \mathbb{C}$, 满足

1. 正定性:

$$\| |v\rangle \| \geq 0$$

且 $\| |v\rangle \| = 0$ iff $|v\rangle = 0_V$.

2. 绝对齐次性:

$$\| c |v\rangle \| = |c| \| |v\rangle \|^$$

3. 三角不等式:

$$\| |v\rangle + |u\rangle \| \leq \| |v\rangle \| + \| |u\rangle \|^$$

则称 $\| \cdot \|$ 为 V 的一个范数 (norm).

Def. 5 赋范空间

若线性空间 V 指定了一个范数 $\| \cdot \|$, 则称 V 为一个赋范空间 (normed space).

Prop. 4

复内积 $\langle \cdot | \cdot \rangle$ 可以诱导出范数

$$\| |v\rangle \| = \sqrt{\langle v | v \rangle}$$

从而复内积空间 V 成为一个赋范空间.

1.2.4 完备性与 Hilbert 空间

Def. 6 序列收敛

设 $\{|v_n\rangle\}_{n=1}^{\infty}$ 为复内积空间 V 中的一个序列. 若存在 $|v\rangle \in V$, 使得: $\forall \epsilon > 0, \exists N \geq \in \mathbb{N}$, s.t. $\forall n \geq N, \| |v_n\rangle - |v\rangle \| < \epsilon$, 则称序列 $\{|v_n\rangle\}$ 收敛 (converge) 到 $|v\rangle$, 记作 $|v_n\rangle \rightarrow |v\rangle$. 或称 $\{|v_n\rangle\}$ 有极限 $|v\rangle$, 记作

$$\lim_{n \rightarrow \infty} |v_n\rangle = |v\rangle \iff \lim_{n \rightarrow \infty} \| |v_n\rangle - |v\rangle \| = 0$$

Def. 7 Cauchy 序列

若 $\forall \epsilon > 0, \exists N \in \mathbb{N}, \text{ s.t. } \forall n, m \geq N, \| |v_n\rangle - |v_m\rangle \| < \epsilon$, 则称 $\{|v_n\rangle\}$ 为 V 中的一个 Cauchy 序列 (Cauchy sequence).

Def. 8 完备性

若 V 中每个 Cauchy 序列均有极限, 则称 V 关于其范数是完备的 (complete).

Def. 9 Hilbert 空间

若内积空间 $\mathcal{H}$ 关于其内积诱导的范数完备, 则称 $\mathcal{H}$ 为一个 Hilbert 空间 (Hilbert space).

Prop. 5

- 有限维复内积空间必定完备, 从而成为一个有限维复 Hilbert 空间.
- 无限维复内积空间不一定完备.

1.2.5 对偶空间**Def. 10 对偶矢量**

设 $\mathcal{H}$ 为一个复 Hilbert 空间. 定义线性映射

$$\langle \phi | : \mathcal{H} \rightarrow \mathbb{C} \quad |\psi\rangle \mapsto \langle \phi | \psi \rangle$$

若 $\forall |\psi\rangle, |\chi\rangle \in \mathcal{H}, a, b, c \in \mathbb{C}$ 满足

1. 保持加法:

$$\langle \phi | (a |\psi\rangle + b |\chi\rangle) = a \langle \phi | \psi \rangle + b \langle \phi | \chi \rangle$$

2. 保持纯量乘法:

$$\langle \phi | (c |\psi\rangle) = c \langle \phi | \psi \rangle$$

则称 $\langle \phi |$ 为 $\mathcal{H}$ 上的一个线性泛函 (linear functional), 或称 $\langle \phi |$ 为矢量 $|\phi\rangle \in \mathcal{H}$ 的对偶矢量 (dual vector).

Rmk 也称矢量 $|\psi\rangle$ 为右矢 (ket), 对偶矢量 $\langle \psi |$ 为左矢 (bra).

Def. 11 对偶空间

$\mathcal{H}$ 中所有矢量的对偶矢量构成的集合为 $\mathbb{C}$ 上的一个复线性空间, 称为 $\mathcal{H}$ 的对偶空间 (dual space).

$$\mathcal{H}^* := \{ \langle \phi | \mid |\phi\rangle \in \mathcal{H} \}$$

1.2.6 矢量的正交与完备

Def. 12 完备正交基

- 设 $\{|u_n\rangle\}$ 为一个矢量集. 若满足

$$\langle u_n | u_m \rangle = \delta_{nm} \quad (1.14)$$

则称 $\{|u_n\rangle\}$ 为正交归一的 (orthonormal).

- 若该矢量集能张成整个 Hilbert 空间, 即该矢量集构成 $\mathcal{H}$ 的一个基, 则称 $\{|u_n\rangle\}$ 为完备的.
- 若 $\{|u_n\rangle\}$ 既是正交归一的, 又是完备的, 则称为一个完备正交基.

Prop. 6 正交归一性条件

- 分立

- 非简并

$$\langle a_n | a_m \rangle = \delta_{nm} \quad (1.15)$$

- 简并

$$\langle a_n^{(i)} | a_m^{(j)} \rangle = \delta_{nm} \delta_{ij} \quad (1.16)$$

- 连续

- 非简并

$$\langle a | a' \rangle = \delta(a - a') \quad (1.17)$$

- 简并

$$\langle a, r | a', r' \rangle = \delta(a - a') \delta_{rr'} \quad (1.18)$$

Prop. 7 完备性条件

- 分立

- 非简并

$$\sum_n |u_n\rangle \langle u_n| = \hat{I} \quad (1.19)$$

- 简并

$$\sum_{n,i} |a_n^{(i)}\rangle \langle a_n^{(i)}| = \sum_n \hat{I}_n = \hat{I} \quad (1.20)$$

- 连续

– 非简并

$$\int da |a\rangle \langle a| = \hat{I} \quad (1.21)$$

– 简并

$$\int da \sum_{r=1}^{g(a)} |a, r\rangle \langle a, r| = \hat{I} \quad (1.22)$$

Pf “ $\Rightarrow$ ”: $\forall |\psi\rangle \in \mathcal{H}, |\psi\rangle = \sum_n c_n |u_n\rangle, \langle u_m|\psi\rangle = \sum_n c_n \underbrace{\langle u_m|u_n\rangle}_{=\delta_{nm}} = c_m,$

$$\Rightarrow |\psi\rangle = \sum_m c_m |u_m\rangle = \sum_m \langle u_m|\psi\rangle |u_m\rangle = \sum_m \underbrace{(|u_m\rangle \langle u_m|)}_{=\hat{I}} |\psi\rangle$$

“ $\Leftarrow$ ”: $|\psi\rangle = \hat{I}|\psi\rangle = \left(\sum_n |u_n\rangle \langle u_n|\right) |\psi\rangle = \sum_n \underbrace{\langle u_n|\psi\rangle}_{=c_n} |u_n\rangle.$

连续情况类似可证.

1.3 Hilbert 空间上的算符

1.3.1 线性算符

Def. 13 线性算符

若 Hilbert 空间上的一个变换 $\hat{A}: \mathcal{H} \rightarrow \mathcal{H}$ 满足

1. 保持加法: $\forall |\psi\rangle, |\phi\rangle \in \mathcal{H},$

$$\hat{A}(|\psi\rangle + |\phi\rangle) = \hat{A}|\psi\rangle + \hat{A}|\phi\rangle$$

2. 保持纯量乘法: $\forall |\psi\rangle \in \mathcal{H}, c \in \mathbb{C}$

$$\hat{A}(c|\psi\rangle) = c\hat{A}|\psi\rangle$$

则称 $\hat{A}$ 为 $\mathcal{H}$ 上的一个线性算符 (linear operator).

- 若 $\forall |\psi\rangle \in \mathcal{H}, \hat{A}|\psi\rangle = \hat{B}|\psi\rangle$, 则称 $\hat{A}$ 与 $\hat{B}$ 相等, 记作 $\hat{A} = \hat{B}$.
- 若 $\forall |\psi\rangle \in \mathcal{H}, \hat{0}|\psi\rangle = 0$, 则称 $\hat{0}$ 为零算符 (zero operator).
- 若 $\forall |\psi\rangle \in \mathcal{H}, \hat{I}|\psi\rangle = |\psi\rangle$, 则称 $\hat{I}$ 为单位算符 (identity operator), 或恒等算符.

Thm. 1 线性算符代数

定义算符的乘法

$$(\hat{A}\hat{B})|\psi\rangle = \hat{A}(\hat{B}|\psi\rangle)$$

于是 $\mathcal{H}$ 上的所有算符, 对于加法、数量乘法、乘法构成一个含么结合代数. 算符的乘法一般非交换

$$\hat{A}\hat{B} = \hat{B}\hat{A} \iff [\hat{A}, \hat{B}] \neq 0$$

1.3.2 对易子**Def. 14 对易子**

设 $\hat{A}, \hat{B}$ 均为 $\mathcal{H}$ 上的线性算符, 称其对易子 (commutator) 为

$$[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A} \quad (1.23)$$

若 $[\hat{A}, \hat{B}] = 0$, 则称 $\hat{A}$ 与 $\hat{B}$ 对易, 否则称其不对易.

Thm. 2 Lie 括号的公理性质

对易子 $[\cdot, \cdot]$ 满足 Lie 括号 (Lie bracket) 的公理化定义:

$$[\cdot, \cdot] : \text{End}(\mathcal{H}) \times \text{End}(\mathcal{H}) \rightarrow \text{End}(\mathcal{H})$$

且 $\forall \hat{A}, \hat{B}, \hat{C} \in \text{End}(\mathcal{H}), a, b \in \mathbb{C}$, 有

1. 双线性

$$\begin{aligned} [a\hat{A} + \hat{B}, \hat{C}] &= a[\hat{A}, \hat{C}] + b[\hat{B}, \hat{C}] \\ [\hat{A}, a\hat{B} + b\hat{C}] &= a[\hat{A}, \hat{B}] + b[\hat{A}, \hat{C}] \end{aligned} \quad (1.24)$$

2. 反对称性

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] \quad (1.25)$$

3. Jacobi 恒等式

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (1.26)$$

从而线性算符空间 $\text{End}(\mathcal{H})$ 在对易子 $[\cdot, \cdot]$ 下构成一个 Lie 代数 (Lie algebra).

Prop. 8 对易子的代数性质

- Leibniz 法则

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \quad [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \quad (1.27)$$

- 幂函数法则: 若 $[\hat{B}, [\hat{A}, \hat{B}]] = 0$, 则

$$[\hat{A}, \hat{B}^n] = n[\hat{A}, \hat{B}]\hat{B}^{n-1} \quad (1.28)$$

- 函数法则: 若 $[\hat{B}, [\hat{A}, \hat{B}]] = 0$, 则

$$[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}] \frac{\partial f}{\partial \hat{B}} \quad (1.29)$$

$$\text{Pf } [\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} + (\hat{B}\hat{A}\hat{C} - \hat{B}\hat{A}\hat{C}) = \underbrace{(\hat{A}\hat{B} - \hat{B}\hat{A})}_{=[\hat{A}, \hat{B}]} \hat{C} + \hat{B} \underbrace{(\hat{A}\hat{C} - \hat{C}\hat{A})}_{=[\hat{A}, \hat{C}]}.$$

对幂指数 n 做数学归纳法: $n = 1$, 显然成立. 设 $n = k$ 成立, 即 $[\hat{A}, \hat{B}^k] = k[\hat{A}, \hat{B}]\hat{B}^{k-1}$. 则对 $n = k + 1$,

$$\begin{aligned} [\hat{A}, \hat{B}^{k+1}] &= \hat{B}[\hat{A}, \hat{B}^k] + [\hat{A}, \hat{B}]\hat{B}^k = k\hat{B}[\hat{A}, \hat{B}]\hat{B}^{k-1} + [\hat{A}, \hat{B}]\hat{B}^k \\ &\stackrel{[\hat{B}, [\hat{A}, \hat{B}]] = 0}{=} k[\hat{A}, \hat{B}]\hat{B}^k + [\hat{A}, \hat{B}]\hat{B}^k = (k+1)[\hat{A}, \hat{B}]\hat{B}^k \end{aligned}$$

命题成立, 从而对于 $n \in \mathbb{N}^*$, 幂函数法则成立.

$$\text{Let } f(\hat{B}) = \sum_{n=0}^{\infty} f_n \hat{B}^n, \text{ then } [\hat{A}, f(\hat{B})] = \left[\hat{A}, \sum_n f_n \hat{B}^n \right] = \sum_n f_n [\hat{A}, \hat{B}^n] = [\hat{A}, \hat{B}] \underbrace{\sum_n n f_n \hat{B}^{n-1}}_{=\frac{\partial f}{\partial \hat{B}}}.$$

1.3.3 本征值与本征矢

Def. 15 本征值, 本征矢, 本征态, 本征空间, 简并度

- 对于 $a \in \mathbb{C}$, 若存在 $|a\rangle \in \mathcal{H} \setminus \{0\}$ 使得 $\hat{A}|a\rangle = a|a\rangle$, 则称 a 为 $\hat{A}$ 的本征值 (eigenvalue), 称 $|a\rangle$ 为 $\hat{A}$ 的属于本征值 a 的本征矢 (eigenvector).
- 与本征矢对应的量子态, 称为本征态 (eigenstate).
- 称由 a 的所有本征矢张成的空间

$$\mathcal{E}_a := \{|a\rangle \in \mathcal{H} \mid \hat{A}|a\rangle = a|a\rangle\} \cup \{0\}$$

为 a 的本征子空间 (eigensubspace).

- 本征空间的维数, 称为本征值 a 的简并度 (degeneracy).

$$g_a := \dim \mathcal{E}_a$$

- 若 $g = 1$, 则称 a 非简并 (non-degenerate).
- 若 $g > 1$, 则称 a 简并 (degenerate).

Def. 16 谱

- 设 $\hat{A}$ 为量子系统的一个可观测量. $\hat{A}$ 的所有本征值构成的集合, 称为 $\hat{A}$ 的本征谱 (eigenspectrum), 记作 $\text{Spec}(\hat{A})$,
- 若可观测量的谱值可以由离散指标记为 $\{a_1, a_2, \dots, a_n, \dots\}$

Def. 17 离散谱与连续谱

1.3.4 外积与投影算符

Def. 18 外积算符

$\forall |\psi\rangle, |\phi\rangle \in \mathcal{H}$, 其外积 $|\psi\rangle\langle\phi|$ 可定义一个线性算符: $\forall |\chi\rangle \in \mathcal{H}$,

$$\left(|\phi\rangle\langle\phi|\right)|\chi\rangle = \left(\langle\phi|\chi\rangle\right)|\psi\rangle \quad (1.30)$$

Def. 19 投影算符

若线性算符 $\hat{P}$ 满足

1. 幂等性 (idempotency)

$$\hat{P}^2 = \hat{P} \quad (1.31)$$

2. 自伴性

$$\hat{P}^\dagger = \hat{P} \quad (1.32)$$

则称 $\hat{P}$ 为一个正交投影算符 (orthogonal projection operator).

若 $|\psi\rangle \in \mathcal{H}$ 归一, 则可定义向 $|\psi\rangle$ 方向的投影算符

$$\hat{P}_\psi = |\psi\rangle\langle\psi| \quad (1.33)$$

易证 $\hat{P}_\psi$ 为一个正交投影算符.

Rmk 只满足幂等性, 而不要求自伴性的算符, 称为投影算符.

Prop. 9 投影算符的本征值问题

投影算符 $\hat{P}_\psi$ 的本征值只有 0, 1 两个取值, 分别对应两个本征矢 $|0\rangle, |1\rangle$. 分别称其为正交态 (orthogonal state) $|\psi_\perp\rangle = |0\rangle$ 与平行态 (parallel state) $|\psi_\parallel\rangle = |1\rangle$.

Prop. 10 任意态矢量的正交投影分解

$\forall |\phi\rangle \in \mathcal{H}, |\phi\rangle = |\phi_\perp\rangle + |\phi_\parallel\rangle$.

$$\begin{cases} |\phi_\perp\rangle = (\hat{I} - \hat{P}_\psi) |\phi\rangle \\ |\phi_\parallel\rangle = \hat{P}_\psi |\phi\rangle \end{cases} \quad (1.34)$$

其中 $|\phi_\perp\rangle, |\phi_\parallel\rangle$ 分别为 $\hat{P}_\psi$ 的属于本征值 0, 1 的本征矢, 即

$$\hat{P}_\psi |\phi_\perp\rangle = 0 \quad \hat{P}_\psi |\phi_\parallel\rangle = |\phi_\parallel\rangle \quad (1.35)$$

$$\text{Pf} \quad \langle \phi_\perp | \phi_\parallel \rangle = \langle \phi | (\hat{I} - \hat{P}_\psi)^\dagger \hat{P}_\psi | \phi \rangle = \langle \phi | \underbrace{(\hat{P}_\psi - \hat{P}_\psi^2)}_{=0} | \phi \rangle = 0$$

Def. 20 本征子空间的投影算符

设 $\hat{A}$ 有简并的本征值 a , 即 $\hat{A} |a^{(i)}\rangle = a |a^{(i)}\rangle, i = 1, \dots, g_a$. 则定义 a 的本征子空间的投影算符为

$$\hat{P}_a = \sum_{i=1}^{g_a} |a^{(i)}\rangle \langle a^{(i)}| \quad (1.36)$$

1.3.5 自伴算符

Def. 21 伴随算符

$\forall |\psi\rangle, |\phi\rangle \in \mathcal{H}$, 有内积关系

$$\langle \phi | \hat{A} | \psi \rangle = \langle \psi | \hat{A}^\dagger | \phi \rangle^* \quad (1.37)$$

称 $\hat{A}^\dagger$ 为线性算符 $\hat{A}$ 的伴随 (adjoint).

Def. 22 自伴算符

若线性算符 $\hat{A}$ 的伴随算符等于其自身, 即

$$\hat{A}^\dagger = \hat{A} \quad (1.38)$$

则称 $\hat{A}$ 为一个自伴算符 (self-adjoint operator).

Rmk 无限维 Hilbert 空间中, 自伴算符还要求定义域

$$D(\hat{A}) = D(\hat{A}^\dagger) \quad (1.39)$$

因此有限维 Hilbert 空间中, 对称算符 $\langle \phi | \hat{A} | \psi \rangle = \langle \psi | \hat{A} | \phi \rangle^*$ 与自伴算符没有分别, 而在无限维 Hilbert 空间中, 可观测量对应的只能是自伴算符.

Thm. 3 自伴算符的本征值与本征矢

自伴算符的本征值必为实值, 且属于不同本征值的本征矢是正交的.

Thm. 4 *

自伴算符的所有本征子空间的直和, 即为 Hilbert 空间. 从而自伴算符的本征矢, 可以构成一个完备正交基.

Prop. 11 算符的自伴-斜自伴分解

满足 $\hat{A}^\dagger = -\hat{A}$ 的算符 $\hat{A}$ 称为斜自伴算符 (skew-self-adjoint operator). 任意一个算符 $\hat{O}$, 均可以分解为自伴算符与斜自伴算符之和.

$$\hat{O} = \frac{1}{2}(\hat{O} + \hat{O}^\dagger) + \frac{1}{2}(\hat{O} - \hat{O}^\dagger) \quad (1.40)$$

其中 $\frac{1}{2}(\hat{O} + \hat{O}^\dagger)$ 为自伴算符, $\frac{1}{2}(\hat{O} - \hat{O}^\dagger)$ 为斜自伴算符.

1.3.6 么正算符

Def. 23 么正算符

若 $\hat{U}$ 为可逆线性算符, 且满足

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{I} \iff \hat{U}^\dagger = \hat{U}^{-1} \quad (1.41)$$

则称 $\hat{U}$ 为么正算符.

Thm. 5 保内积性质

么正算符不改变态矢量之间的内积. 设 $|\psi\rangle \rightarrow |\psi'\rangle = \hat{U} |\psi\rangle$, $|\phi\rangle \rightarrow |\phi'\rangle = \hat{U} |\phi\rangle$. 则

$$\langle \phi' | \psi' \rangle = \langle \phi | \hat{U}^\dagger \hat{U} | \psi \rangle \quad (1.42)$$

Prop. 12 等效算符变换

若态矢量进行主动变换 $|\psi'\rangle = \hat{U}|\psi\rangle$, 可观测量不变, 则可将态矢量上的变换转移到算符上:

$$\hat{A}_{\text{eff}} = \hat{U}^\dagger \hat{A} \hat{U} \quad (1.43)$$

在等效下, 可观测量 $\hat{A}$ 的测量期望值不变.

$$\langle \psi' | \hat{A} | \psi' \rangle = \langle \psi | \underbrace{\hat{U}^\dagger \hat{A} \hat{U}}_{=\hat{A}_{\text{eff}}} | \psi \rangle \quad (1.44)$$

Rmk

Prop. 13 算符的主动变换

$$\hat{A}' = \hat{U} \hat{A} \hat{U}^\dagger \quad (1.45)$$

Prop. 14 么正算符的本征值

对于 $\hat{U} |u\rangle = u |u\rangle$, $|u\rangle \neq 0$, 有

$$|u| = 1 \quad u = e^{i\theta} \quad (\theta \in \mathbb{R}) \quad (1.46)$$

即, 么正算符的本征值位于复平面的单位圆上.

Thm. 6 么正群

Hilbert 空间 $\mathcal{H}$ 上所有么正算符对于映射的复合构成一个群, 称为么正群 (unitary group).

$$\mathbf{U}(\mathcal{H}) := \{ \hat{U} \in \text{GL}(\mathcal{H}) \mid \hat{U}^\dagger = \hat{U}^{-1} \} \quad (1.47)$$

$\mathbf{U}(\mathcal{H})$ 中所有行列式为 1 的算符也构成一个群, 称为特殊么正群 (special unitary group).

$$\mathbf{SU}(\mathcal{H}) := \{ \hat{U} \in \mathbf{U}(\mathcal{H}) \mid \det \hat{U} = 1 \} \quad (1.48)$$

Thm. 7 三性合一

以下算符的三种性质中, 任意两个成立时, 第三个必成立:

1. 自伴性: $\hat{A}^\dagger = \hat{A}$;
2. 么正性: $\hat{A}^\dagger = \hat{A}^{-1}$;
3. 对合性: $\hat{A}^2 = \hat{I}$.

此时算符的本征值必为 1 或 -1.

Pf $\hat{A}^2 |a\rangle = a\hat{A} |a\rangle = a^2 |a\rangle = |a\rangle, \implies a \in \{\pm 1\}.$

1.4 表象

Def. 24

在 Hilbert 空间中选定一个完备正交基, 称由此基建立的一个坐标系, 为一个表象 (representation). 可观测量 $\hat{A}$ 的本征矢 $\{|a\rangle\}$ 构成一个完备正交基, 从而建立 $\hat{A}$ -表象, 或称 $\{|a\rangle\}$ -表象.

1.4.1 波函数

Def. 25 波函数

态矢量在一个表象下的展开系数, 称为波函数 (wavefunction).

Prop. 15 分立表象下的波函数

在 $\{|a_n\rangle\}$ -表象下, 态矢量 $|\psi\rangle$ 展开为

$$|\psi\rangle = \sum_n \underbrace{\langle a_n | \psi \rangle}_{=c_n} |a_n\rangle \quad (1.49)$$

其中 $c_n = \langle a_n | \psi \rangle$ 称为 $|\psi\rangle$ 在 $\{|a_n\rangle\}$ -表象下的波函数的第 n 个分量.

Prop. 16 连续表象下的波函数

在 $\{|a\rangle\}$ -表象下, 态矢量 $|\psi\rangle$ 展开为

$$|\psi\rangle = \int da \underbrace{\langle a | \psi \rangle}_{=c(a)} |a\rangle \quad (1.50)$$

其中 $c(a) = \langle a | \psi \rangle$ 称为 $|\psi\rangle$ 在 $\{|a\rangle\}$ -表象下的波函数.

1.4.2 算符在表象下的矩阵表示

Thm. 8 算符的矩阵表示

选定表象 $\{|u_n\rangle\}$, 算符 $\hat{A}$ 在该表象下的矩阵表示为

$$\hat{A}(|u_1\rangle, |u_2\rangle, \dots, |u_n\rangle, \dots) = (|u_1\rangle, |u_2\rangle, \dots, |u_n\rangle, \dots) \mathbf{A} \quad (1.51)$$

$\mathbf{A}$ 的矩阵元为

$$A_{nm} = \langle u_n | \hat{A} | u_m \rangle \quad (1.52)$$

Pf 将算符写为外积展开的形式

$$\hat{A} = \hat{I} \hat{A} \hat{I} = \left(\sum_n |u_n\rangle \langle u_n| \right) \hat{A} \left(\sum_m |u_m\rangle \langle u_m| \right) = \sum_{nm} \underbrace{\langle u_n | \hat{A} | u_m \rangle}_{=A_{nm}} |u_n\rangle \langle u_m|$$

Thm. 9 *

非简并算符 $\hat{A}$ 在自身表象下的矩阵为对角矩阵

$$\hat{A} = \text{diag}(a_1, \dots, a_n) \quad (1.53)$$

Def. 26 算符的迹

算符在一个表象下的矩阵的迹, 称为算符的迹 (trace).

$$\text{Pf} \quad \text{tr} \hat{A} = \sum_{n,m} A_{nm} \underbrace{\text{tr}(|u_n\rangle \langle u_m|)}_{=\delta_{nm}} = \sum_n A_{nn}.$$

Prop. 17 迹的性质

$$\text{tr}(|\psi\rangle \langle \phi|) = \langle \phi | \psi \rangle \quad (1.54)$$

$$\text{tr}(\hat{A} \hat{B}) = \text{tr}(\hat{B} \hat{A}) \quad (1.55)$$

$$\text{tr}(\hat{A}_1 \hat{A}_2 \cdots \hat{A}_n) = \text{tr}(\hat{A}_2 \cdots \hat{A}_n \hat{A}_1) = \text{tr}(\cdots \hat{A}_n \hat{A}_1 \hat{A}_2) \quad (1.56)$$

Prop. 18 伴随的矩阵表示

在一个表象下, $\mathbf{A}^\dagger$ 为 $\mathbf{A}$ 的复共轭转置矩阵 (complex conjugate transpose)

$$(\mathbf{A}^\dagger)_{ij} = A_{ji}^* \quad (1.57)$$

1.4.3 表象变换

Thm. 10 基矢变换

基 $\{u_n\}$ 到 $\{v_n\}$ 的变换为

$$|v_n\rangle = \hat{S} |u_n\rangle \quad (1.58)$$

其中 $\hat{S}$ 称为表象变换算符 (representation transformation operator).

$$\hat{S} = \sum_m |v_m\rangle \langle u_m| \quad (1.59)$$

$\hat{S}$ 与 $\hat{S}^\dagger$ 的矩阵元分别为

$$\langle u_k | \hat{S} | u_l \rangle = \langle v_k | \hat{S} | v_l \rangle = \langle u_k | v_l \rangle \quad (1.60)$$

$$\langle u_k | \hat{S}^\dagger | u_l \rangle = \langle v_k | \hat{S}^\dagger | v_l \rangle = \langle v_k | u_l \rangle \quad (1.61)$$

$$\text{Pf} \quad |v_n\rangle = \sum_m |u_m\rangle \underbrace{\langle u_m | v_n \rangle}_{:=S_{mn}} = \sum_m u_m \langle u_m | \hat{S} | u_n \rangle = \underbrace{\left(\sum_m |u_m\rangle \langle u_m| \right)}_{=\hat{I}} \hat{S} |u_n\rangle.$$

Thm. 11 表象变换算符

表象变换算符 $\hat{S}$ 是个么正算符

$$\hat{S}^\dagger = \hat{S}^{-1} \iff \hat{S}^\dagger \hat{S} = \hat{S} \hat{S}^\dagger = \hat{I} \quad (1.62)$$

Rmk 表象变换不改变 Hilbert 空间的度量性质, 从而不改变量子系统的物理内容, 只改变数学描述的方式.

Cor. 1 逆变换

$$|u_n\rangle = \hat{S}^\dagger |v_n\rangle \quad (1.63)$$

Prop. 19 分量变换

设 $|\psi\rangle = \sum_n c_n |u_n\rangle = \sum_m d_m |v_m\rangle$, $c_n = \langle u_n | \psi \rangle$, $d_m = \langle v_m | \psi \rangle$. 态矢量的分量变换为

$$d_m = \sum_n S_{nm}^* c_n = \sum_n (\mathbf{S}^\dagger)_{mn} c_n \iff \mathbf{d} = \mathbf{S}^\dagger \mathbf{c} \quad (1.64)$$

Prop. 20 算符变换

$$\hat{A}_v = \hat{S}^\dagger \hat{A}_u \hat{S} \quad (1.65)$$

$$\text{Pf} \quad (\hat{A}_v)_{kl} = \langle v_k | \hat{A} | v_l \rangle, (\hat{A}_u)_{kl} = \langle u_k | \hat{A} | u_l \rangle.$$

$$\begin{aligned} \langle v_k | \hat{A} | v_l \rangle &= \sum_{n,m} \langle v_k | u_n \rangle \langle u_n | \hat{A} | u_m \rangle \langle u_m | v_l \rangle = \sum_{n,m} \langle u_k | \hat{S}^\dagger | u_n \rangle \langle u_n | \hat{A} | u_m \rangle \langle u_m | \hat{S} | u_l \rangle \\ &= \langle u_k | \hat{S}^\dagger \left(\sum_n |u_n\rangle \langle u_n| \right) \hat{A} \left(\sum_m |u_m\rangle \langle u_m| \right) \hat{S} | u_l \rangle = \langle u_k | \hat{S}^\dagger \hat{A} \hat{S} | u_l \rangle \end{aligned}$$

1.4.4 位置表象与动量表象

Thm. 12 $\hat{x}$ 与 $\hat{p}$ 在 $\hat{x}$ -表象下的表示

$$\hat{x} = \mathbf{x} \quad \hat{p} = -i\hbar \nabla \quad (1.66)$$

$$\hat{x}_i = x_i \quad \hat{p}_i = -i\hbar \frac{\partial}{\partial x_i} \quad (1.67)$$

Prop. 21 位置表象的本征方程与完备正交性条件

$$\hat{x} |\mathbf{x}\rangle = \mathbf{x} |\mathbf{x}\rangle \quad (1.68)$$

$$\langle \mathbf{x} | \mathbf{x}' \rangle = \delta(\mathbf{x} - \mathbf{x}') \quad (1.69)$$

$$\int d^3x |\mathbf{x}\rangle \langle \mathbf{x}| = \hat{I} \quad (1.70)$$

Prop. 22 动量表象的完备正交性条件

$$\hat{p} |\mathbf{p}\rangle = \mathbf{p} |\mathbf{p}\rangle \quad (1.71)$$

$$\langle \mathbf{p} | \mathbf{p}' \rangle = \delta(\mathbf{p} - \mathbf{p}') \quad (1.72)$$

$$\int d^3p |\mathbf{p}\rangle \langle \mathbf{p}| = \hat{I} \quad (1.73)$$

Thm. 13 位置表象与动量表象下的波函数

$$\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle \quad (1.74)$$

$$\tilde{\psi}(\mathbf{p}) = \langle \mathbf{p} | \psi \rangle \quad (1.75)$$

Thm. 14 表象变换

1.5 可观测量算符

1.5.1 可观测量的对易关系

Post. 6 基本对易关系

将经典力学基本 Poisson 括号 $[x_i, x_j]_{\text{PB}} = [p_i, p_j]_{\text{PB}} = 0$, $[x_i, p_j]_{\text{PB}} = \delta_{ij}$ 经正则量子化替换

$$[A, B]_{\text{PB}} \longleftrightarrow \frac{1}{i\hbar} [\hat{A}, \hat{B}] \quad (1.76)$$

可得量子力学的基本对易关系

$$[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0 \quad (1.77)$$

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}\hat{I} \quad (1.78)$$

1.5.2 相容可观测量与量子数

Def. 27 相容可观测量

若可观测量 $\hat{A}, \hat{B}$ 对应的算符对易, 即

$$[\hat{A}, \hat{B}] = 0 \quad (1.79)$$

则称 A 与 B 为相容可观测量.

Thm. 15 相容可观测量定理

$[\hat{A}, \hat{B}] = 0 \iff \hat{A}$ 与 $\hat{B}$ 有共同本征态集合, 且其构成一个正交完备集.

Pf

1.5.3 期望值与方差

Def. 28 期望值

可观测量 $\hat{A}$ 的期望值 (expectation value) 为

$$\langle \hat{A} \rangle \stackrel{\text{norm.}}{=} \langle \psi | \hat{A} | \psi \rangle \stackrel{\text{unnorm.}}{=} \frac{\langle \psi | \hat{A} | \psi \rangle}{\langle \psi | \psi \rangle} \quad (1.80)$$

Rmk $\langle \hat{A} \rangle$ 由 $|\psi\rangle$ 决定, 反映了系统的量子态.

$$\mathbf{Pf} \quad \langle \hat{A} \rangle = \sum_n P(a_n) a_n = \sum_n \langle \psi | a_n \rangle \langle a_n | \psi \rangle a_n = \langle \psi | \left(\sum_n |a_n\rangle \langle a_n| \hat{A} \right) | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle.$$

Def. 29 均方差

可观测量 $\hat{A}$ 的均方差 (variance) 为

$$(\Delta A)^2 = \left\langle \left(\hat{A} - \langle \hat{A} \rangle \right)^2 \right\rangle = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \quad (1.81)$$

标准差 (stand deviation) 或称均方根 (root mean square, rms) 为

$$\Delta A = \sqrt{(\Delta A)^2} = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2} \quad (1.82)$$

Rmk 均方差更细致地反映了系统的量子态.

1.5.4 不确定性关系

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (1.83)$$

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle| \quad (1.84)$$

1.5.5 纯态与混合态

pure state mixed state

1.5.6 密度算符

若由于缺乏系统的信息,导致无法确定其状态,而该种不确定性并非由量子力学的内禀不确定性导致,从而可用系综描述这种经典统计力学的不确定性.

Def. 30 密度算符

若系统处在 $|\psi_i\rangle$ 态的经典概率为 p_i , 则可定义密度算符 (density operator)

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad (1.85)$$

若系统处在 $|\psi(\lambda)\rangle$ 态的经典概率密度为 $f(\lambda)$, 则密度算符为

$$\hat{\rho} = \int d\lambda f(\lambda) |\psi(\lambda)\rangle \langle \psi(\lambda)| \quad (1.86)$$

Thm. 16 密度算符形式的期望值公式

$$\langle \hat{A} \rangle_\rho = \text{tr}(\hat{\rho} \hat{A}) \quad (1.87)$$

Prop. 23 闭系统的条件

若 $\sum_i p_i = 1$ 或 $\int d\lambda f(\lambda) = 1$, 则称系统为闭系统 (closed system), 条件为

$$\text{tr} \hat{\rho} = 1 \quad (1.88)$$

Prop. 24 密度算符的性质

1. 么正性

$$\hat{\rho}^\dagger = \hat{\rho} \quad (1.89)$$

2. 纯度: $\text{tr}(\hat{\rho}^2) \leq 1$, 当且仅当 $\hat{\rho} = |\psi\rangle\langle\psi|$, 即系统为纯态时取等. 对于混合态

$$\text{tr}(\hat{\rho}^2) < 1 \quad (1.90)$$

3. 若 $\hat{\rho} = \sum_{nm} \rho_{nm} |a_n\rangle\langle a_m|$, 则

$$\rho_{nm} = \rho_{mn}^* \quad (1.91)$$

若 $\hat{\rho} = \int da da' \rho(a, a') |a\rangle\langle a'|$, 则

$$\rho(a, a') = \rho(a', a)^* \quad (1.92)$$

1.6 张量积

2 量子动力学

Def. 1 绘景

在描述量子系统的时间演化时, 将时间依赖性如何分配给态矢量和算符, 且保持物理预言不变的约定, 称为绘景 (picture).

2.1 Schrödinger 绘景

Def. 2 时间演化算符

时间演化算符 $\hat{U}$ 是一个把系统从初始时刻 t_0 的态矢量, 么正地演化到任意时刻 t 的态矢量的线性算符.

$$|\psi(t, t_0)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle \quad (2.1)$$

Thm. 1 时间演化算符的 Schrödinger 方程

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H} \hat{U}(t, t_0) \quad (2.2)$$

$$\hat{U}(t, t_0) = e^{-\frac{i}{\hbar} \hat{H}(t-t_0)} \quad (2.3)$$

$$\hat{U}(t, t_0) = \exp\left(-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt'\right) \quad (2.4)$$

Thm. 2 Dyson 级数

$$\hat{U}(t, t_0) = 1 + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n \hat{H}(t_1) \hat{H}(t_2) \cdots \hat{H}(t_n) \quad (2.5)$$

Def. 3 Heisenberg 绘景下的可观测量算符

$$\hat{A}_H(t) = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t) \quad (2.6)$$

2.2 Heisenberg 绘景

Thm. 3 Heisenberg 方程

$$\frac{d}{dt} \hat{A} = \frac{1}{i\hbar} [\hat{A}_H, \hat{H}] \quad (2.7)$$

2.3 相互作用绘景

2.4 Feynman 路径积分

Thm. 4 Feynman 路径积分公式

$$\langle x_N, t_N | x_1, t_1 \rangle = \int_{x_1}^{x_N} \mathcal{D}[x(t)] \exp\left(\frac{i}{\hbar} \int_{t_1}^{t_N} dt L(x, \dot{x})\right) \quad (2.8)$$

2.5 密度算符的时间演化

3 角动量

Thm. 1 角动量的基本对易关系

$$[\hat{J}_i, \hat{J}_j] = i\hbar \varepsilon_{ijk} \hat{J}_k \quad (3.1)$$

3.1 Pauli 矩阵

Def. 1 Pauli 矩阵

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.2)$$

Thm. 2 Pauli 矩阵基定理

$\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ 构成 $\mathbb{C}^{2 \times 2}$ 的一个基. 其中

$$\sigma_0 = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3.3)$$

从而任意 2 阶复矩阵 $\mathbf{A}$, 均可由 $\{\sigma_\mu\}_{\mu=0,1,2,3}$ 展开:

$$\mathbf{A} = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix} = x_0 \sigma_0 + \vec{x} \cdot \vec{\sigma} \quad (3.4)$$

其中 $\vec{x} \cdot \vec{\sigma} = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3$.

$$\text{Pf} \quad \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow x_0 = \frac{a+d}{2}, x_1 = \frac{b+c}{2}, x_2 = i \frac{b-c}{2}, x_3 = \frac{a-d}{2}.$$

Prop. 1

$$\sigma_i \sigma_j = \delta_{ij} + i \varepsilon_{ijk} \sigma_k \quad (3.5)$$

$$\text{tr } \sigma_i = 0 \quad (3.6)$$

4 双态系统

$$\hat{\rho} = a \hat{\sigma}_0 + \vec{b} \cdot \hat{\sigma}$$

$$\text{tr } \hat{\rho} = 2a \quad (4.1)$$

$$\langle \hat{\sigma} \rangle = \text{tr}(\hat{\rho} \hat{\sigma}_i) = 2b_i \quad (4.2)$$

$$\hat{S}_i = \frac{\hbar}{2} \hat{\sigma}_i \quad (4.3)$$

$$\Rightarrow \hat{\rho} = \frac{1}{2} \left(\hat{I} + \frac{2}{\hbar} \langle \hat{S} \rangle \cdot \hat{\sigma} \right) \quad (4.4)$$

Prop. 1 特殊么正算符的 Pauli 算符分解

2 阶特殊么正算符 $\hat{U} \in \text{SO}(2)$ 可以分解为 Pauli 算符 $\{\hat{\sigma}_\mu\}_{\mu=0,1,2,3}$.

$$\hat{U} = x_0 \hat{\sigma}_0 + i \vec{x} \cdot \hat{\sigma} \quad (x_\mu \in \mathbb{R}) \quad (4.5)$$

Prop. 2 自伴算符与斜自伴算符的 Pauli 算符分解

2 阶自伴算符 $\hat{A}$ 可以分解为 Pauli 算符 $\{\hat{\sigma}_\mu\}_{\mu=0,1,2,3}$.

$$\hat{A} = x_0 \hat{\sigma}_0 + \vec{x} \cdot \hat{\sigma} \quad (x_\mu \in \mathbb{R}) \quad (4.6)$$

2 阶斜自伴算符 $\hat{B}$ 也可以分解为

$$\hat{B} = i y_0 \hat{\sigma}_0 + i \vec{y} \cdot \hat{\sigma} \quad (y_\mu \in \mathbb{R}) \quad (4.7)$$

5 对称性

5.1 连续对称性与守恒律

5.1.1 空间平移与动量守恒

Def. 1 空间平移算符

量子力学中的空间平移, 由空间平移算符 (space translation operator) 描述, 若将位置本征态移动 $\mathbf{a}$:

$$\hat{T}(\mathbf{a})|\mathbf{x}\rangle := |\mathbf{x} + \mathbf{a}\rangle \quad (5.1)$$

Rmk $\langle \mathbf{x} | \hat{T}(\mathbf{a})^\dagger = \langle \mathbf{x} + \mathbf{a} |$.

Thm. 1 Abel 群性质

设所有空间平移算符构成的集合为 $\mathcal{T}(3) := \{\hat{T}(\mathbf{a}) \mid \mathbf{a} \in \mathbb{R}^3\}$, 在其中定义乘法

$$\hat{T}(\mathbf{a})\hat{T}(\mathbf{b}) = \hat{T}(\mathbf{b} + \mathbf{a}) \quad (5.2)$$

空间平移算符的乘法满足以下群公理: $\forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$

1. 结合律:

$$[\hat{T}(\mathbf{a})\hat{T}(\mathbf{b})]\hat{T}(\mathbf{c}) = \hat{T}(\mathbf{a})[\hat{T}(\mathbf{b})\hat{T}(\mathbf{c})] \quad (5.3)$$

2. 有单位元:

$$\hat{T}(\mathbf{0}) = \hat{I} \quad (5.4)$$

3. 有逆元:

$$\hat{T}(\mathbf{a})^{-1} := \hat{T}(-\mathbf{a}) \quad (5.5)$$

以及乘法交换律

$$\hat{T}(\mathbf{a})\hat{T}(\mathbf{b}) = \hat{T}(\mathbf{b})\hat{T}(\mathbf{a}) \quad (5.6)$$

从而 $\mathcal{T}(3)$ 构成一个 Abel 群.

Pf 乘法封闭性: $[\hat{T}(\mathbf{a})\hat{T}(\mathbf{b})]|\mathbf{x}\rangle = \hat{T}(\mathbf{a})|\mathbf{x} + \mathbf{b}\rangle = |\mathbf{x} + \mathbf{b} + \mathbf{a}\rangle = \hat{T}(\mathbf{b} + \mathbf{a})|\mathbf{x}\rangle$.

结合律: $[\hat{T}(\mathbf{a})\hat{T}(\mathbf{b})]\hat{T}(\mathbf{c}) = \hat{T}(\mathbf{b} + \mathbf{a})\hat{T}(\mathbf{c}) = \hat{T}(\mathbf{c} + \mathbf{b} + \mathbf{a}) = \hat{T}(\mathbf{a})\hat{T}(\mathbf{c} + \mathbf{b}) = \hat{T}(\mathbf{a})[\hat{T}(\mathbf{b})\hat{T}(\mathbf{c})]$.

单位元: $\hat{T}(\mathbf{0})\hat{T}(\mathbf{a}) = \hat{T}(\mathbf{a})\hat{T}(\mathbf{0}) = \hat{T}(\mathbf{a})$.

逆元: $\hat{T}(\mathbf{a})^{-1}\hat{T}(\mathbf{a}) = \hat{T}(-\mathbf{a})\hat{T}(\mathbf{a}) = \hat{T}(\mathbf{a} - \mathbf{a}) = \hat{T}(\mathbf{0})$. 同理, $\hat{T}(\mathbf{a})\hat{T}(\mathbf{a})^{-1} = \hat{T}(\mathbf{0})$.

交换律: $[\hat{T}(\mathbf{a})\hat{T}(\mathbf{b})]|\mathbf{x}\rangle = |\mathbf{x} + \mathbf{b} + \mathbf{a}\rangle = |\mathbf{x} + \mathbf{a} + \mathbf{b}\rangle = [\hat{T}(\mathbf{b})\hat{T}(\mathbf{a})]|\mathbf{x}\rangle$.

Prop. 1 空间平移算符的么正性

空间平移算符为么正算符, 即

$$\hat{T}(\mathbf{a})^\dagger \hat{T}(\mathbf{a}) = \hat{T}(\mathbf{a}) \hat{T}(\mathbf{a})^\dagger = \hat{I} \iff \hat{T}(\mathbf{a})^\dagger = \hat{T}(\mathbf{a})^{-1} \quad (5.7)$$

Pf Let $|\psi'\rangle = \hat{T}(\mathbf{a})|\psi\rangle$, $|\phi'\rangle = \hat{T}(\mathbf{a})|\phi\rangle$, then

$$\begin{aligned} \langle\phi'|\psi'\rangle &= \langle\phi|\hat{T}(\mathbf{a})^\dagger \hat{T}(\mathbf{a})|\psi\rangle = \langle\phi|\left[\int d^3x |\mathbf{x}\rangle\langle\mathbf{x}|\right] \hat{T}(\mathbf{a})^\dagger \hat{T}(\mathbf{a}) \left[\int d^3x' |\mathbf{x}'\rangle\langle\mathbf{x}'|\right]|\psi\rangle \\ &= \int d^3x \int d^3x' \underbrace{\langle\phi|\mathbf{x}\rangle\langle\mathbf{x}+\mathbf{a}|\mathbf{a}'+\mathbf{a}\rangle}_{=\delta(\mathbf{x}-\mathbf{x}')} \underbrace{\langle\mathbf{x}|\psi\rangle}_{=\hat{I}} = \langle\phi|\left(\int d^3x |\mathbf{x}\rangle\langle\mathbf{x}|\right)|\psi\rangle \end{aligned}$$

从而 $\hat{T}(\mathbf{a})^\dagger \hat{T}(\mathbf{a}) = \hat{I}$, 再由乘法交换律, 得证.

Cor. 1 对左矢的作用

$$\langle\mathbf{x}|\hat{T}(\mathbf{a}) = \langle\mathbf{x}-\mathbf{a}| \quad (5.8)$$

Pf $\langle\mathbf{x}|\hat{T}(\mathbf{a}) = \langle\mathbf{x}|\left[\hat{T}(\mathbf{a})^{-1}\right]^\dagger = \langle\mathbf{x}|\left[\hat{T}(-\mathbf{a})\right]^\dagger = \langle\mathbf{x}-\mathbf{a}|$.

Prop. 2 对波函数的作用

$$\hat{T}(\mathbf{x})\psi(\mathbf{x}) = \psi(\mathbf{x}-\mathbf{a}) \quad (5.9)$$

Pf $\hat{T}(\mathbf{a})\psi(\mathbf{x}) = \langle\mathbf{x}|\hat{T}(\mathbf{a})|\psi\rangle = \langle\mathbf{x}-\mathbf{a}|\psi\rangle = \psi(\mathbf{x}-\mathbf{a})$.

Prop. 3 与位置算符的对易关系

$$[\hat{\mathbf{x}}, \hat{T}(\mathbf{a})] = \mathbf{a}\hat{T}(\mathbf{a}) \quad (5.10)$$

Pf $[\hat{\mathbf{x}}, \hat{T}(\mathbf{a})]|\psi\rangle = [\hat{\mathbf{x}}, \hat{T}(\mathbf{a})]\left(\int d^3x |\mathbf{x}\rangle\langle\mathbf{x}|\right)|\psi\rangle = \int d^3x [\hat{\mathbf{x}}, \hat{T}(\mathbf{a})]|\mathbf{x}\rangle\langle\mathbf{x}|\psi\rangle$.
 $[\hat{\mathbf{x}}, \hat{T}(\mathbf{a})]|\mathbf{x}\rangle = \hat{\mathbf{x}}\hat{T}(\mathbf{a})|\mathbf{x}\rangle - \hat{T}(\mathbf{a})\hat{\mathbf{x}}|\mathbf{x}\rangle$

Prop. 4 位置算符的主动变换

$$\hat{\mathbf{x}} \rightarrow \hat{\mathbf{x}}' = \hat{T}(\mathbf{a})\hat{\mathbf{x}}\hat{T}(\mathbf{a})^\dagger = \hat{\mathbf{x}} - \mathbf{a}\hat{I} \quad (5.11)$$

Pf $[\hat{\mathbf{x}}, \hat{T}(\mathbf{a})] = \mathbf{a}\hat{T}(\mathbf{a}) \implies \hat{T}(\mathbf{a})\hat{\mathbf{x}}\hat{T}(\mathbf{a})^\dagger = [\hat{\mathbf{x}}\hat{T}(\mathbf{a}) - \mathbf{a}\hat{T}(\mathbf{a})]\hat{T}(\mathbf{a})^\dagger = \hat{\mathbf{x}} - \mathbf{a}\hat{I}$.

5.1.2 空间旋转与轨道角动量守恒

5.1.3 时间平移与能量守恒

5.2 空间反演与宇称

5.2.1 宇称算符

Def. 2 宇称算符

量子力学中的空间反演, 由宇称算符 (parity operator) 描述:

$$\hat{\Pi} |\mathbf{x}\rangle = |-\mathbf{x}\rangle \quad (5.12)$$

Prop. 5 宇称算符的性质

宇称算符自伴、么正且对合, 即

$$\hat{\Pi}^\dagger = \hat{\Pi}^{-1} = \hat{\Pi} \quad \hat{\Pi}^2 = \hat{I} \quad (5.13)$$

三性合一, 从而其本征值 $\pi \in \{\pm 1\}$.

$$\text{Pf} \quad \hat{\Pi}^2 |\mathbf{x}\rangle = \hat{\Pi} |-\mathbf{x}\rangle = |\mathbf{x}\rangle. \quad \forall |\psi\rangle \in \mathcal{H}, \quad \hat{\Pi}^2 |\psi\rangle = \int d^3x \langle \mathbf{x} | \psi \rangle \underbrace{\hat{\Pi}^2 |\mathbf{x}\rangle}_{=|\mathbf{x}\rangle} = |\psi\rangle.$$

Def. 3 奇偶宇称

称宇称算符 $\hat{\Pi}$ 的本征值 $\pi \in \{\pm 1\}$ 为宇称 (parity). 称 $\pi_+ = 1$ 为偶宇称 (even parity), 称 $\pi_- = -1$ 为奇宇称 (odd parity).

Cor. 2 动量本征态的宇称

$$\hat{\Pi} |\mathbf{p}\rangle = |-\mathbf{p}\rangle \quad (5.14)$$

5.2.2 奇偶宇称态

Def. 4 奇偶宇称态

宇称算符 $\hat{\Pi}$ 的属于本征值 $\pi = 1$ 的本征态, 称为偶宇称态 (even parity state); 属于 $\pi = -1$ 的本征态, 称为奇宇称态 (odd parity state).

$$\hat{\Pi} |\psi_+\rangle = |\psi_+\rangle \quad \hat{\Pi} |\psi_-\rangle = -|\psi_-\rangle \quad (5.15)$$

Prop. 6 位置表象下的波函数

奇偶宇称态的波函数是奇偶函数, 即

$$\psi_+(-\mathbf{x}) = \psi_+(\mathbf{x}) \quad \psi_-(-\mathbf{x}) = -\psi_-(\mathbf{x}) \quad (5.16)$$

$$\text{Pf} \quad \langle \mathbf{x} | \hat{\Pi} | \psi_+ \rangle = \begin{cases} \langle -\mathbf{x} | \psi_+ \rangle = \psi_+(-\mathbf{x}) \\ \langle \mathbf{x} | \psi_+ \rangle = \psi_+(\mathbf{x}) \end{cases} \implies \psi_+(-\mathbf{x}) = \psi_+(\mathbf{x}).$$

$$\langle \mathbf{x} | \hat{\Pi} | \psi_- \rangle = \begin{cases} \langle -\mathbf{x} | \psi_- \rangle = \psi_-(-\mathbf{x}) \\ -\langle \mathbf{x} | \psi_- \rangle = -\psi_-(\mathbf{x}) \end{cases} \implies \psi_-(-\mathbf{x}) = -\psi_-(\mathbf{x}).$$

Def. 5 奇偶宇称空间

偶宇称 $\pi_+ = 1$ 的本征子空间

$$\mathcal{H}_+ := \{ |\pi_+ \rangle \in \mathcal{H} \mid \Pi |\pi_+ \rangle = |\pi_+ \rangle \} \quad (5.17)$$

称为偶宇称子空间 (even parity subspace).

奇宇称 $\pi_- = -1$ 的本征子空间

$$\mathcal{H}_- := \{ |\pi_- \rangle \in \mathcal{H} \mid \Pi |\pi_- \rangle = -|\pi_- \rangle \} \quad (5.18)$$

称为奇宇称子空间 (odd parity subspace).

Thm. 2

整个 Hilbert 空间可正交分解为偶宇称子空间与奇宇称子空间的直和.

$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_- \quad (5.19)$$

Pf 由宇称算符 $\hat{\Pi}$ 的自伴性可得.

5.2.3 奇偶投影算符

Def. 6 奇偶投影算符

偶投影算符 (even projection operator) $\hat{P}_+$ 与奇投影算符 (odd projection operator) $\hat{P}_-$ 定义为

$$\hat{P}_{\pm} = \frac{1}{2} (\hat{I} \pm \hat{\Pi}) \quad (5.20)$$

其满足幂等性 $\hat{P}_{\pm}^2 = \hat{P}_{\pm}$ 与自伴性 $\hat{P}_{\pm}^\dagger = \hat{P}_{\pm}$, 从而是正交投影算符.

$$\text{Pf} \quad \hat{P}_{\pm}^2 = \frac{1}{4} (\hat{I} \pm 2\hat{\Pi} + \hat{\Pi}^2) = \frac{1}{4} (2\hat{I} \pm 2\hat{\Pi}) = \hat{P}_{\pm}.$$

$$\hat{P}_{\pm}^{\dagger} = \frac{1}{2} (\hat{I} \pm \hat{\Pi})^{\dagger} = \frac{1}{2} (\hat{I} \pm \hat{\Pi}^{\dagger}) = \hat{P}_{\pm}.$$

Prop. 7 奇偶投影算符的性质

- 完备性

$$\hat{P}_{+} + \hat{P}_{-} = \hat{I} \quad (5.21)$$

- 正交性

$$\hat{P}_{+}\hat{P}_{-} = \hat{P}_{-}\hat{P}_{+} = 0 \quad (5.22)$$

- 与宇称算符的对易性

$$[\hat{P}_{\pm}, \hat{\Pi}] = 0 \quad (5.23)$$

Prop. 8 奇偶投影

宇称算符 $\hat{\Pi}$ 有本征方程

$$\hat{\Pi} (\hat{P}_{\pm} |\psi\rangle) = \pm \hat{P}_{\pm} |\psi\rangle \quad (5.24)$$

从而 $\hat{P}_{\pm} |\psi\rangle$ 为 $\hat{\Pi}$ 的属于本征值 ± 1 的本征态.

奇偶投影算符可将任意态 $|\psi\rangle$ 变换为奇偶宇称态

$$\hat{P}_{\pm} |\psi\rangle = |\psi_{\pm}\rangle \quad (5.25)$$

Prop. 9 态矢量的奇偶分解

$\forall |\psi\rangle \in \mathcal{H}$, 有唯一的奇偶正交分解

$$|\psi\rangle = |\psi_{+}\rangle + |\psi_{-}\rangle \quad (5.26)$$

其中 $|\psi_{+}\rangle \in \mathcal{H}_{+}$, $|\psi_{-}\rangle \in \mathcal{H}_{-}$. 且有

$$\langle \psi_{+} | \psi_{-} \rangle = 0 \quad (5.27)$$

$$\text{Pf } |\psi\rangle = \hat{P}_{+} |\psi\rangle + (\hat{I} - \hat{P}_{+}) |\psi\rangle = \underbrace{\hat{P}_{+} |\psi\rangle}_{=|\psi_{+}\rangle} + \underbrace{(\hat{I} - \hat{P}_{+}) |\psi\rangle}_{=|\psi_{-}\rangle}.$$

5.2.4 奇偶算符
Def. 7 偶算符

若算符 $\hat{A}_{+}$ 与宇称算符 $\hat{\Pi}$ 对易, 即满足

$$[\hat{\Pi}, \hat{A}_{+}] = 0 \quad (5.28)$$

则称 $\hat{A}_+$ 为一个偶算符 (even operator).

Prop. 10 偶算符的宇称性质

偶算符在宇称变换下不变.

$$\hat{A}_+ = \hat{\Pi} \hat{A}_+ \hat{\Pi}^\dagger \quad (5.29)$$

$$\text{Pf } [\hat{\Pi}, \hat{A}_+] = 0, \implies \hat{\Pi} \hat{A}_+ = \hat{A}_+ \hat{\Pi}, \implies \hat{A}_+ = \hat{\Pi} \hat{A}_+ \hat{\Pi}^\dagger.$$

Def. 8 奇算符

若算符 $\hat{A}_-$ 与宇称算符 $\hat{\Pi}$ 反对易, 即满足

$$\{\hat{\Pi}, \hat{A}_-\} = 0 \quad (5.30)$$

则称 $\hat{A}_-$ 为一个奇算符 (odd operator).

Prop. 11 奇算符的宇称性质

奇算符在宇称变换下符号相反.

$$\hat{A}_- = -\hat{\Pi} \hat{A}_- \hat{\Pi}^\dagger \quad (5.31)$$

$$\text{Pf } \{\hat{\Pi}, \hat{A}_-\} = 0, \implies \hat{\Pi} \hat{A}_- = -\hat{A}_- \hat{\Pi}. \implies \hat{A}_- = -\hat{\Pi} \hat{A}_- \hat{\Pi}^\dagger.$$

Thm. 3 算符的奇偶分解

若 $\hat{\Pi} \hat{A} \hat{\Pi}^\dagger = \eta_A \hat{A}$, 其中 $\eta_A \in \{\pm 1\}$, 则称 $\hat{A}$ 有确定宇称. 而一般算符不具有确定宇称. 任意算符均可以分解为偶算符与奇算符之和 $\hat{A} = \hat{A}_+ + \hat{A}_-$.

$$\hat{A}_+ = \frac{1}{2} (\hat{A} + \hat{\Pi} \hat{A} \hat{\Pi}^\dagger) \quad \hat{A}_- = \frac{1}{2} (\hat{A} - \hat{\Pi} \hat{A} \hat{\Pi}^\dagger) \quad (5.32)$$

Thm. 4 奇偶算符的乘法运算

设 $\hat{A}_+, \hat{B}_+$ 与 $\hat{A}_-, \hat{B}_-$ 为任意偶算符与奇算符, 则

算符乘积	奇偶性
$\hat{A}_+ \hat{B}_+$	偶算符
$\hat{A}_+ \hat{B}_-$	奇算符
$\hat{A}_- \hat{B}_+$	奇算符
$\hat{A}_- \hat{B}_-$	偶算符

$$\text{Pf } \text{Let } \pi, \pi' \in \{\pm 1\}, \text{ then } \hat{\Pi} \hat{A}_\pi \hat{\Pi}^\dagger = \pi \hat{A}_\pi \text{ and } \hat{\Pi} \hat{B}_{\pi'} \hat{\Pi}^\dagger = \pi' \hat{B}_{\pi'}.$$

$$\hat{\Pi} \hat{A}_\pi \hat{B}_{\pi'} \hat{\Pi}^\dagger = \hat{\Pi} \hat{A}_\pi \hat{\Pi}^\dagger \hat{\Pi} \hat{B}_{\pi'} \hat{\Pi}^\dagger = \pi \pi' \hat{A}_\pi \hat{B}_{\pi'}$$

Prop. 12 对易子与反对易子的宇称性质

设 $\pi, \pi' \in \{\pm 1\}$, 则 $[\hat{A}_\pi, \hat{B}_{\pi'}]$ 与 $\{\hat{A}_\pi, \hat{B}_{\pi'}\}$ 的宇称均为 $\pi\pi'$. 有

$$\hat{\Pi}[\hat{A}_\pi, \hat{B}_{\pi'}]\hat{\Pi}^\dagger = \pi\pi'[\hat{A}_\pi, \hat{B}_{\pi'}] \quad \hat{\Pi}\{\hat{A}_\pi, \hat{B}_{\pi'}\}\hat{\Pi}^\dagger = \pi\pi'\{\hat{A}_\pi, \hat{B}_{\pi'}\} \quad (5.33)$$

Prop. 13 算符函数的宇称性质

若 $f(\hat{A}) = \sum_n f_n \hat{A}^n$, 则

- $f(\hat{A}_+)$ 为偶算符;
- 当 $f(\hat{A})$ 为偶函数时, $f(\hat{A}_-)$ 为偶算符; $f(\hat{A})$ 为奇函数时, $f(\hat{A}_-)$ 为奇算符.

Pf $\hat{\Pi}f(\hat{A}_+)\hat{\Pi}^\dagger = \sum_n f_n \hat{\Pi} \hat{A}_+^n \hat{\Pi}^\dagger = \sum_n f_n \underbrace{(\hat{\Pi} \hat{A}_+^n \hat{\Pi}^\dagger)}_n = \sum_n f_n \hat{A}_+^n = f(\hat{A}_+).$

$$\hat{\Pi}f(\hat{A}_-)\hat{\Pi}^\dagger = \sum_n f_n \hat{\Pi} \hat{A}_-^n \hat{\Pi}^\dagger = \sum_n (-1)^n f_n \hat{A}_-^n \neq f(\hat{A}_-).$$

If $f(\hat{A})$ even, $\hat{\Pi}f(\hat{A}_-)\hat{\Pi}^\dagger = \sum_n f_n \hat{A}_-^n = f(\hat{A}_-)$, then $f(\hat{A}_-)$ even.

If $f(\hat{A})$ odd, $\hat{\Pi}f(\hat{A}_-)\hat{\Pi}^\dagger = -\sum_n f_n \hat{A}_-^n = -f(\hat{A}_-)$, then $f(\hat{A}_-)$ odd.

Prop. 14 宇称的保持与翻转 *

偶算符保持宇称, 奇算符翻转宇称.

Eg. 1 $\hat{x}, \hat{p}, \hat{L}$ 的奇偶性

位置算符与动量算符均为奇算符.

$$\hat{\Pi}\hat{x}\hat{\Pi}^\dagger = -\hat{x} \quad (5.34)$$

$$\hat{\Pi}\hat{p}\hat{\Pi}^\dagger = -\hat{p} \quad (5.35)$$

轨道角动量算符为偶算符

$$\hat{\Pi}\hat{L}\hat{\Pi}^\dagger = \hat{L} \quad (5.36)$$

Pf $\hat{\Pi}\hat{L}_i\hat{\Pi}^\dagger = \hat{\Pi}\varepsilon_{ijk}\hat{x}_j\hat{p}_k\hat{\Pi}^\dagger = \varepsilon_{ijk}\hat{\Pi}\hat{x}_j\hat{\Pi}^\dagger\hat{\Pi}\hat{p}_k\hat{\Pi}^\dagger = \varepsilon_{ijk}(-\hat{x}_j)(-\hat{p}_k) = \hat{L}_i.$

Def. 9 极矢量与轴矢量

- 满足 $\{\hat{\Pi}, \hat{v}_i\} = 0$, 或 $\hat{\Pi}\hat{v}_i\hat{\Pi}^\dagger = -\hat{v}_i$ 的矢量算符, 即奇矢量算符, 对应的矢量 $\mathbf{v}$ 称为极矢量 (polar vector), 或称真矢量 (true vector).
- 满足 $[\hat{\Pi}, \hat{v}_i] = 0$, 或 $\hat{\Pi}\hat{v}_i\hat{\Pi}^\dagger = \hat{v}_i$ 的矢量算符, 即偶矢量算符, 对应的矢量 $\mathbf{v}$ 称为轴矢量 (axial vector), 或称赝矢量 (pseudo vector).

Def. 10 真标量与赝标量

- 奇标量算符 $\hat{Q}$ 对应的标量 Q 称为标量, 或者真标量 (true scalar).
- 偶标量算符 $\hat{Q}$ 对应的矢量 Q 称为赝标量 (pseudo scalar).

5.2.5 与宇称相关的跃迁

Thm. 5 宇称选择定则

设跃迁算符 $\hat{A}$ 有确定宇称 η_A , 初态 $|i\rangle$ 与末态 $|f\rangle$ 的宇称分别为 π_i, π_f . 则要求

$$\pi_f \eta_A \pi_i = 1 \quad (5.37)$$

- 对于偶算符 $\hat{A}_+$, 要求 $\pi_f \pi_i = 1$, 即相同宇称的态允许, 相反宇称的态禁戒.
- 对于奇算符 $\hat{A}_-$, 要求 $\pi_f \pi_i = -1$, 即相反宇称的态允许, 相同宇称的态禁戒.

称该与宇称相关的跃迁规则为宇称选择定则 (parity selection rule), 跃迁能否发生, 还需结合其他选择定则.

$$\text{Pf} \quad \langle f | \hat{A} | i \rangle = \langle f | \hat{\Pi}^\dagger \underbrace{\hat{\Pi} \hat{A} \hat{\Pi}^\dagger}_{=\eta_A \hat{A}} \hat{\Pi} | i \rangle = \pi_f \eta_A \pi_i \langle f | \hat{A} | i \rangle, \implies \pi_f \eta_A \pi_i = 1.$$

Eg. 2 电偶极跃迁

电偶极矩算符为

$$\hat{\mathbf{d}} = \sum_n q_n \hat{\mathbf{x}}_n \quad (5.38)$$

由于 $\hat{\mathbf{x}}_n$ 为极矢量算符, 故 $\hat{\mathbf{d}}$ 为奇算符, 因此电偶极跃迁要求 $\pi_f = -\pi_i$, 即初末态宇称相反.

Eg. 3 磁偶极跃迁

磁偶极矩算符为

$$\hat{\boldsymbol{\mu}} = -\mu_B \sum_n (\hat{\mathbf{L}}_n + g_s \hat{\mathbf{S}}_n) \quad (5.39)$$

由于 $\hat{\mathbf{L}}_n, \hat{\mathbf{S}}_n$ 均为轴矢量算符, 故 $\hat{\boldsymbol{\mu}}$ 为偶算符, 因此磁偶极跃迁要求 $\pi_f = \pi_i$, 即初末宇称相同.

5.3 时间反演

5.3.1 反么正算符

Def. 11 反线性算符

定义算符 $\hat{\Theta} : \mathcal{H} \rightarrow \mathcal{H}$, 若满足 $\forall |\psi\rangle, |\phi\rangle \in \mathcal{H}, a, b \in \mathbb{C}$,

$$\hat{\Theta}(a|\psi\rangle, b|\phi\rangle) = a^* (\hat{\Theta}|\psi\rangle) + b^* \hat{\Theta}|\phi\rangle \quad (5.40)$$

则称 $\hat{\Theta}$ 为一个反线性算符 (anti-linear operator).

Prop. 15 么正-反么正算符的乘法规则

设 $\hat{U}, \hat{U}_1, \hat{U}_2$ 均为么正算符, $\hat{\Theta}, \hat{\Theta}_1, \hat{\Theta}_2$ 均为反么正算符.

- $\hat{U}_1 \hat{U}_2$ 为么正算符.
- $\hat{U} \hat{\Theta}$ 为反么正算符.
- $\hat{\Theta} \hat{U}$ 为反么正算符.
- $\hat{\Theta}_1 \hat{\Theta}_2$ 为么正算符.

6 GuoHong's Lecture

6.1 Fundamental Postulates of QM

1. Hilbert Space

- (a) Non-relativistic QM
- (b) Closed system
- (c) pure state

fundamental postulates:

SOME IS

2. Group Theory and Symmetry & Conservation

(a) Discrete Group

- i. Space Inversion $\leftrightarrow \hat{I}$ (Parity is conserved, except in weak interactions.)
- ii. Time Reversal $\leftrightarrow$ Kramers degeneracy

(b) Continuous group (Lie group)

- i. Space translation $\leftrightarrow \mathbf{p}$
- ii. Space rotations $\leftrightarrow \mathbf{J}$
- iii. Time translation $\leftrightarrow E$

3. Toolbox

(a) Physics: CM (Lagrangian and Hamiltonian Mechanics), CED, CTD, SM.

- i. Euler-Lagrangian eq.

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^\alpha} = \frac{\partial L}{\partial q^\alpha} \quad \text{vs} \quad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = \frac{\partial \mathcal{L}}{\partial \phi}$$

- ii. Lagrangian-based QM: path integral.
- iii. Lagrangian is Lorentz invariant (scalar), Hamiltonian is NOT Lorentz invariant.
- iv. Canonical coordinates: position q and momentum p .
- v. Phase space: phase point $\rightarrow$ state.
- vi. Canonical commutators: Poisson bracket

$$[A, B]_{\text{PB}} \longleftrightarrow \frac{1}{i\hbar} [\hat{A}, \hat{B}]$$

$$\text{Eg. } [x_i, p_j]_{\text{PB}} = \delta_{ij} \longleftrightarrow [\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}.$$

vii. Canonical eqs of motion (general)

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [f, H]_{\text{PB}} \quad \text{vs} \quad \begin{cases} \frac{d\hat{F}_H}{dt} = \left(\frac{\partial \hat{F}_S}{\partial t} \right)_H + \frac{1}{i\hbar} [\hat{F}_H, \hat{H}] & \text{(Heisenberg eq)} \\ \frac{d}{dt} \langle \hat{F} \rangle = \left\langle \frac{\partial \hat{F}}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle [\hat{F}, H] \rangle & \text{(Ehrenfest thm)} \end{cases}$$

viii. Canonical Transformation

ix. Canonical Quantization: $A(q, p, t) \rightarrow \hat{A}(\hat{q}, \hat{p}, t)$, where $[\hat{q}_i, \hat{p}_j] = i\hbar\delta_{ij}$.

(b) Mathematics: Vector Analysis, Functional Analysis, Group Theory.

4. Fundamental Postulates

(a) State postulate

$$\begin{cases} |\psi\rangle & \text{statevector} \in \mathcal{H} \text{ (ket)} \\ \langle\psi| = |\psi\rangle^\dagger & \text{statevector} \in \mathcal{H}^* \text{ (bra, dual vector)} \end{cases}$$

(b) Operator (Observable) postulate

$\hat{A}$: linear Hermitian operator for observable.

(c) Measurement postulate

$$\hat{A} = |\psi_n\rangle = A_n |\psi_n\rangle$$

$$P(A_n) = |\langle\psi_n|\psi\rangle|^2$$

$$|\psi\rangle \xrightarrow{\text{measurement}} |\psi_n\rangle$$

(d) Evolution (dynamics) postulate

i. Schrödinger Picture

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = \hat{H} |\psi(t)\rangle$$

ii. Heisenberg picture

$$\frac{d\hat{F}_H}{dt} = \left(\frac{\partial \hat{F}_S}{\partial t} \right)_H + \frac{1}{i\hbar} [\hat{F}_H, \hat{H}]$$

(e) Identical Particles and Symmetrization

i. Bosons: $s = 0, 1, 2, \dots$ (integer) Bose-Einstein statistical distribution

ii. Fermions: $s = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$ (half-integer) Fermi-Dirac statistical distribution

- Vectors in Hilbert Space

$\mathcal{H}$: countable/uncountable, linear space (vector space), inner product space.

– vector Addition

* Commutativity: $\forall |\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$,

$$|\psi_1\rangle + |\psi_2\rangle = |\psi_2\rangle + |\psi_1\rangle$$

* Associativity: $\forall |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle \in \mathcal{H}$,

$$(|\psi_1\rangle + |\psi_2\rangle) + |\psi_3\rangle = |\psi_1\rangle + (|\psi_2\rangle + |\psi_3\rangle)$$

* Zero: $\exists 0 \in \mathcal{H}$, s.t. $\forall |\psi\rangle \in \mathcal{H}$,

$$0 + |\psi\rangle = |\psi\rangle + 0 = |\psi\rangle$$

* Inverse: $\forall |\psi\rangle \in \mathcal{H}$, $\exists (-|\psi\rangle) \in \mathcal{H}$,

$$|\psi\rangle + (-|\psi\rangle) = (-|\psi\rangle) + |\psi\rangle = 0$$

– Scalar Multiplication: $\forall |\psi\rangle \in \mathcal{H}, c \in \mathbb{C}, c|\psi\rangle \in \mathcal{H}$.

* Distributivity over scalar addition: $\forall c_1, c_2 \in \mathbb{C}, |\psi\rangle \in \mathcal{H}$,

$$(c_1 + c_2)|\psi\rangle = c_1|\psi\rangle + c_2|\psi\rangle$$

* Distributivity over vector addition: $\forall c \in \mathbb{C}, |\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$,

$$c(|\psi_1\rangle + |\psi_2\rangle) = c|\psi_1\rangle + c|\psi_2\rangle$$

* Associativity: $\forall c_1, c_2 \in \mathbb{C}, |\psi\rangle \in \mathcal{H}$,

$$(c_1 c_2)|\psi\rangle = c_1(c_2|\psi\rangle)$$

* Unit: $\exists 1 \in \mathbb{C}$

– Inner Product: $\forall |\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}, (|\psi_1\rangle, |\psi_2\rangle) \in \mathbb{C}$.

* Hermitian

$$(|\psi_1\rangle, |\psi_2\rangle) = (|\psi_2\rangle, |\psi_1\rangle)^*$$

* Linear in the Second Argument

$$(|\psi\rangle, c_1|\psi_1\rangle + c_2|\psi_2\rangle) = c_1(|\psi\rangle, |\psi_1\rangle) + c_2(|\psi\rangle, |\psi_2\rangle)$$

* Conjugate-linear in the First Argument

$$(c_1|\psi_1\rangle + c_2|\psi_2\rangle, |\psi\rangle) = c_1^*(|\psi\rangle, |\psi_1\rangle) + c_2^*(|\psi\rangle, |\psi_2\rangle)$$

- * Positive Definiteness (norm)

$$\| |\psi\rangle \| := (|\psi\rangle, |\psi\rangle) \geq 0 \quad \text{equality iff } |\psi\rangle = 0$$

represented by adjoint

$$(|\psi_1\rangle, |\psi_2\rangle) = |\psi_1\rangle^\dagger |\psi_2\rangle = \langle \psi_1 | \psi_2 \rangle$$

- Operator in Hilbert Space (Transformation)

$$|\psi\rangle \xrightarrow{\hat{A}} |\psi'\rangle \iff \hat{A} |\psi\rangle = |\psi'\rangle$$

- Linear Operators: $\forall c_1, c_2 \in \mathbb{C}, |\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$,

$$\hat{A}(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle) = c_1 \hat{A} |\psi_1\rangle + c_2 \hat{A} |\psi_2\rangle$$

.

- * $\hat{A} = \hat{B}$: $\forall |\psi\rangle \in \mathcal{H}, \hat{A} |\psi\rangle = \hat{B} |\psi\rangle$.
- * Operator addition: $(\hat{A} + \hat{B}) |\psi\rangle = \hat{A} |\psi\rangle + \hat{B} |\psi\rangle$.
- * Operator multiplication: $(\hat{A}\hat{B}) |\psi\rangle = \hat{A}(\hat{B} |\psi\rangle)$. Generally

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}$$

- Adjoint: Hermitian Conjugation

$$(\langle \psi | \hat{A})^\dagger = \hat{A}^\dagger | \psi \rangle \iff \langle \phi | \hat{A}^\dagger | \psi \rangle = \langle \psi | \hat{A} | \phi \rangle^*$$

- * $(\hat{A}^\dagger)^\dagger = \hat{A}$
- * $(c\hat{A})^\dagger = c^* \hat{A}^\dagger$
- * $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$
- * $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

- Outer Product: $|\psi\rangle \langle \phi|$ is an operator.

$$(|\psi\rangle \langle \phi|) |\chi\rangle = (\langle \phi | \chi \rangle) |\psi\rangle$$

Def. 1 Hermitian Operator (Self-adjoint Operator)

$$\hat{A}^\dagger = \hat{A}$$

Def. 2 Unitary Operator

$$\hat{U}^\dagger = \hat{U}^{-1} \iff \hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{I}$$

Def. 3 Anti-unitary Operator

Anti-unitary Operator is NOT unitary.

Duality, Trinity, Quaternity

1. Hermitian

$$\hat{A}^\dagger = \hat{A}$$

2. Unitary

$$\hat{U}^\dagger = \hat{U}^{-1}$$

3. Involutory

$$\hat{A}^2 = \hat{I}$$

4. Idempotent

$$\hat{A}^2 = \hat{A}$$

Prop. 1 Trinity

A Hermitian unitary operator is necessarily involutory.

Rmk Generally, if an operator satisfies any two of the three properties (Hermiticity, unitarity, involution), it must satisfy the third.

$$\text{Pf } \hat{A}^\dagger = \hat{A} = \hat{A}^{-1}, \implies \hat{A}^2 = \hat{A}\hat{A} = \hat{A}^{-1}\hat{A} = \hat{A}\hat{A}^{-1} = \hat{I}.$$

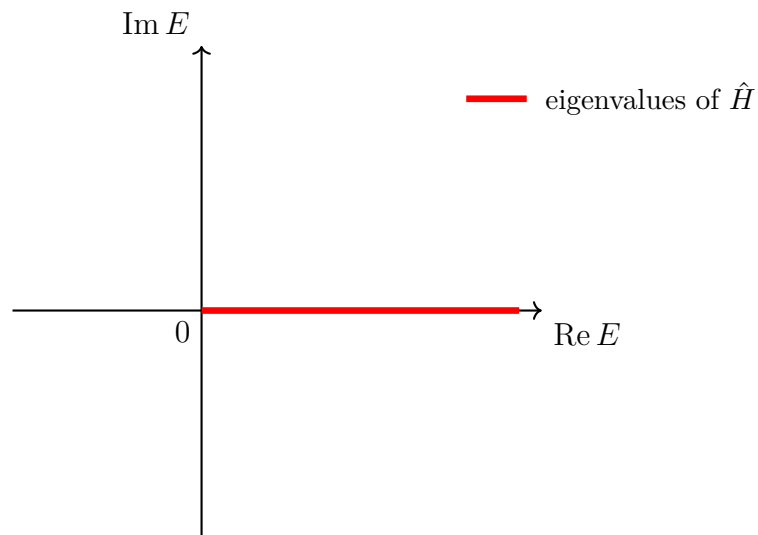
6.2 Eigenvalue Problem**6.2.1 Eigenvalue and Eigenstate**

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle \quad (\lambda \in \mathbb{C})$$

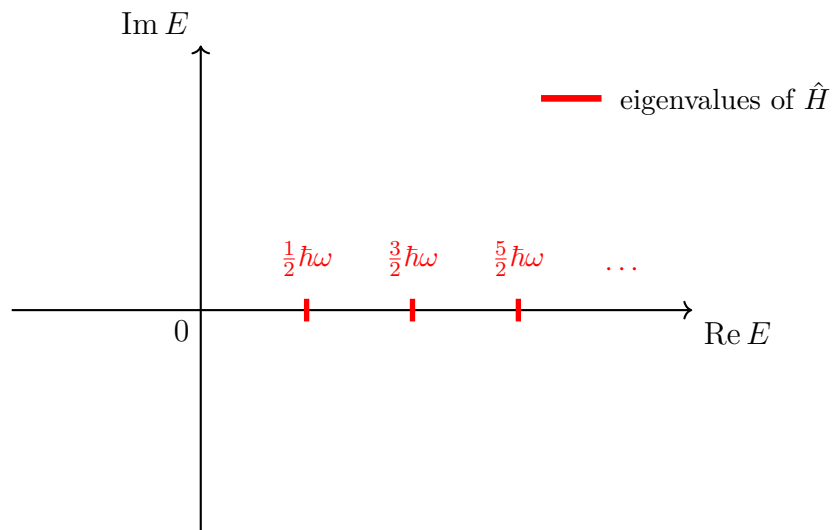
1. Hermitian operator (self-adjoint) $\hat{A}^\dagger = \hat{A}$, observable;
2. Unitary operator $\hat{A}^\dagger = \hat{A}^{-1}$ or $\hat{A}^\dagger \hat{A} = \hat{A} \hat{A}^{-1} = \hat{I}$, symmetric transformation, except for time reversal.

examples of eigenvalues

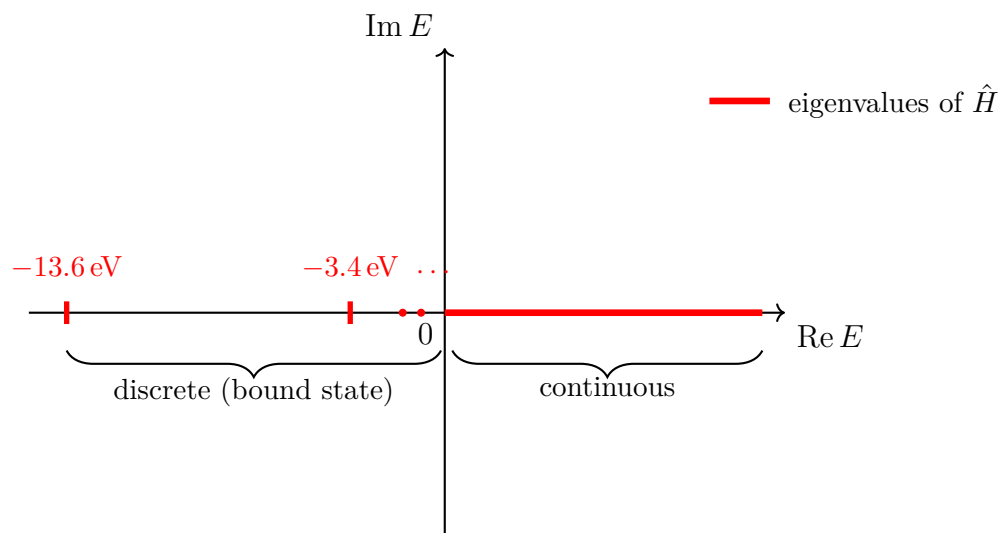
1. free particle: $\hat{H} = \frac{\hat{p}^2}{2m}$ (continuous)



2. harmonic oscillator: $\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2 x^2$ (discrete)



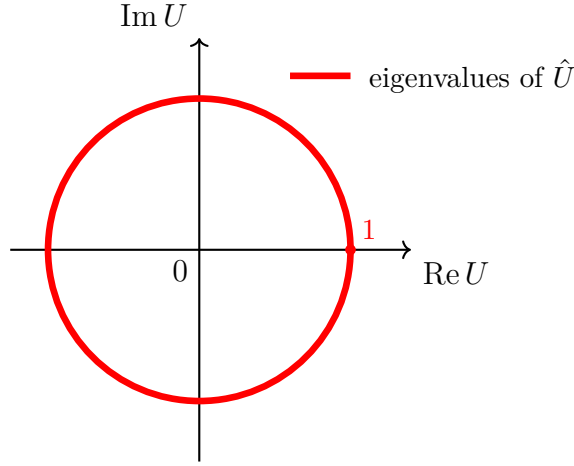
3. hydrogen atom: $\hat{H} = \frac{\hat{p}^2}{2m} - \frac{e^2}{r}$



4. Unitary operator: Let $\hat{U} |u\rangle = u |u\rangle$, $\Rightarrow \langle u | \hat{U} = \langle u | u^*$,

$$\Rightarrow \langle u | \hat{U}^\dagger \hat{U} |u\rangle = \langle u | u^* u |u\rangle = |u|^2 \langle u | u\rangle = \langle u | \hat{I} |u\rangle = \langle u | u\rangle$$

$$\Rightarrow |u|^2 = 1, \Rightarrow u = e^{i\theta} \in \mathbb{C}.$$



$$\hat{U}^\dagger |u\rangle = e^{-i\theta} |u\rangle$$

Eigenvalues of Hermitian operators are $\in \mathbb{R}$, and Unitary operators are $\in \mathbb{C}$. The eigenvalues of an operator that is both Hermitian and unitary are $\in \{\pm 1\}$.

Exercises: Proof thms

1. If $\hat{A}^\dagger = \hat{A}$, $\hat{B}^\dagger = \hat{B}$, then $(\hat{A}\hat{B})^\dagger = \hat{A}\hat{B}$ iff $[\hat{A}, \hat{B}] = 0$.

$$\text{Pf } (\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger = \hat{B}\hat{A} \stackrel{[\hat{A}, \hat{B}]=0}{=} \hat{A}\hat{B}.$$

2. If $\hat{A}^\dagger = \hat{A}^{-1}$, $\hat{B}^\dagger = \hat{B}^{-1}$, then $(\hat{A}\hat{B})^\dagger = (\hat{A}\hat{B})^{-1}$.

$$\text{Pf } (\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger = \hat{B}^{-1} \hat{A}^{-1} = (\hat{A}\hat{B})^{-1}.$$

3. If $\hat{A}$ hermitian, then its eigenvalues are real.

Rmk Let $\hat{A} |\psi\rangle = a |\psi\rangle$, then $\langle \psi | \hat{A} |\psi\rangle = a \langle \psi | \psi\rangle = \langle \psi | \hat{A} |\psi\rangle^* = a^* \langle \psi | \psi\rangle$, $\Rightarrow a = a^*$, $a \in \mathbb{R}$.

4. If $\hat{A}$ is hermitian, then its eigenstates corresponding to distinct eigenvalues are orthogonal. (non-degenerate case)

$$\text{Pf } \text{Let } \hat{A} |\psi\rangle = a |\psi\rangle, \hat{A} |\psi'\rangle = a' |\psi'\rangle.$$

$$\langle \psi' | \hat{A} |\psi\rangle = a \langle \psi' | \psi\rangle = \langle \psi' | \hat{A}^\dagger |\psi\rangle = (\langle \psi' | \hat{A} |\psi'\rangle) = (a' \langle \psi' | \psi'\rangle) = a'^* \langle \psi' | \psi\rangle$$

$\Rightarrow (a - a') \langle \psi' | \psi\rangle = 0$, $\stackrel{a \neq a'}{\Rightarrow} \langle \psi' | \psi\rangle = 0$, i.e. $|\psi\rangle \perp |\psi'\rangle$. In the degenerate case, one must resort to the Gram-Schmidt orthogonalization procedure.

Thm. 1

Eigenstates of a hermitian operator can form an orthogonal, normalized, complete basis.

- discrete basis (non-degenerate): $\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$.
 - orthonormal: $\langle\psi_n|\psi_m\rangle = \delta_{nm}$
 - complete (meaning the basis is sufficient to span the Hilbert space):

$$\sum_n |\psi_n\rangle\langle\psi_n| = \hat{I} \text{ (resolution of identity)}$$

Rmk δ_{nm} is a symmetric tensor.

- continuous basis: $\hat{A}|a\rangle = a|a\rangle$
 - orthonormal: $\langle a|a'\rangle = \delta(a - a')$
 - complete: $\int |a\rangle\langle a| da = \hat{I}$

Rmk $\delta(a - a')$ is an even function.

6.2.2 Eigenspectrum

Eigenspectrum is the set of eigenvalues.

example of eigenspectrum

1. free particle: $\text{Spec}(\hat{H}) = \{a \mid a \in \mathbb{R}, a > 0\}$
2. harmonic oscillator: $\text{Spec}(\hat{H}) = \left\{\left(n + \frac{1}{2}\right)\hbar\omega \mid n = 0, 1, 2, \dots\right\}$.
3. H atom: $\text{Spec}(\hat{H}) = \left\{\frac{1}{n^2}E_g \mid n = 0, 1, 2, \dots\right\} \cup \{a \mid a \in \mathbb{R}, a > 0\}$. Mix spectrum.

Rmk ON: $\langle n, l, m | n', l', m' \rangle = \delta_{nn'}\delta_{ll'}\delta_{mm'}$, $\langle E, l, m | E', l', m' \rangle = \delta(E - E')\delta_{ll'}\delta_{mm'}$.

RI: $\sum_{n,l,m} |n, l, m\rangle\langle n, l, m| + \int dE \sum_{l,m} |E, l, m\rangle\langle E, l, m| = \hat{I}$.

4. Unitary operator: $\text{Spec}(\hat{U}) = \{u \in \mathbb{C} \mid |u| = 1\}$

6.2.3 Eigenspace

Degenerate case (one eigenvalue corresponds to multiple eigenstates):

$$\hat{A}|\psi_n^i\rangle = a_n|\psi_n^i\rangle \quad (i = 1, 2, \dots, g_n)$$

where g_n denotes the degeneracy.

Def. 4 Subspace

$$\mathcal{H}_n = \text{Span} \left\{ |\psi_n^i\rangle \mid i = 1, 2, \dots, g_n \right\}$$

$$\mathcal{H} = \bigoplus_{n=1}^N \mathcal{H}_n$$

$$\dim \mathcal{H}_n = g_n, \dim \mathcal{H} = \dim \bigoplus_{n=1}^N \mathcal{H}_n = \sum_{n=1}^N \dim \mathcal{H}_n = \sum_{n=1}^N g_n.$$

Direct Sum is for the same observable operator.

$$[\hat{\mathbf{L}}^2, \hat{L}_z] = 0, \text{ so } \hat{\mathbf{L}}^2 \text{ \& } \hat{L}_z \text{ have the same eigenstate } |l, m_l\rangle. \hat{\mathbf{L}}^2 |l, m_l\rangle = (l+1)\hbar^2 |l, m_l\rangle.$$

$$\begin{cases} l = 0, m_l = 0 & \rightarrow |0, 0\rangle \quad (1 \text{ dim}) \\ l = 1, m_l = 0, \pm 1 & \rightarrow |1, -1\rangle, |1, 0\rangle, |1, 1\rangle \quad (3 \text{ dim}) \\ \vdots & \end{cases}$$

$$\dim \mathcal{H}_{\hat{\mathbf{L}}^2} = \sum_{l=0}^{l_{\max}} (2l+1)$$

For bound states, the upper limit $l_{\max}$ is finite and determined by n (eg. $l_{\max} = n-1$ in H).

For unbound states, $l_{\max} \rightarrow \infty$.

$$\text{ON: } \langle \psi_n^i | \psi_{n'}^{i'} \rangle = \delta_{nn'} \delta_{ii'}, \text{ RI: } \sum_{n=1}^N \sum_{i=1}^{g_n} |\psi_n^i\rangle \langle \psi_n^i| = \sum_{n=1}^N \hat{I}_n = \hat{I}.$$

Gram-Schmidt Orthogonalization

$$\begin{cases} |\phi_n^1\rangle = |\chi_n^1\rangle = \frac{|\psi_n^1\rangle}{\sqrt{\langle \psi_n^1 | \psi_n^1 \rangle}} \\ |\chi_n^2\rangle = |\psi_n^2\rangle - \langle \phi_n^1 | \psi_n^2 \rangle |\phi_n^1\rangle & \rightarrow |\phi_n^2\rangle = \frac{|\chi_n^2\rangle}{\sqrt{\langle \chi_n^2 | \chi_n^2 \rangle}} \\ \vdots & \vdots \\ |\chi_n^{g_n}\rangle = |\psi_n^{g_n}\rangle - \sum_{j=1}^{g_n-1} \langle \phi_n^j | \psi_n^{g_n} \rangle |\phi_n^j\rangle & \rightarrow |\phi_n^{g_n}\rangle = \frac{|\chi_n^{g_n}\rangle}{\sqrt{\langle \chi_n^{g_n} | \chi_n^{g_n} \rangle}} \end{cases}$$

6.3 Tensor Product

Origin: multi-freedom.

1. Multi particles: s_1, s_2 ;
2. Single particle, but multi observables ($\hat{\mathbf{L}}^2, \hat{L}_z, \hat{\mathbf{S}}^2, \hat{S}_z$)
3. Single particle, single observable, but multi components ($\hat{\mathbf{x}} \rightarrow \hat{x}, \hat{y}, \hat{z}$)

Def. 5 Tensor Product

$|\psi\rangle_1 \in \mathcal{H}_1$, $|\phi\rangle_2 \in \mathcal{H}_2$, then $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, and $|\chi\rangle = |\psi\rangle_1 \otimes |\phi\rangle_2 \in \mathcal{H}$.

6.3.1 Properties of tensor product

The subscript “ i ” indicates that the vector $|\psi\rangle_i$ belongs to the subspace $\mathcal{H}_i$.

1. Commutativity

$$|\psi\rangle_1 \otimes |\phi\rangle_2 = |\phi\rangle_2 \otimes |\psi\rangle_1$$

2. Linear: $\forall c \in \mathbb{C}$, $|\psi\rangle_1 \in \mathcal{H}_1$, $|\phi\rangle_2 \in \mathcal{H}_2$,

$$c(|\psi\rangle_1 \otimes |\phi\rangle_2) = (c|\psi\rangle_1) \otimes |\phi\rangle_2 = |\psi\rangle_1 \otimes (c|\phi\rangle_2)$$

3. Distributivity

$$(a|\psi_1\rangle_1 + b|\psi_2\rangle_1) \otimes |\phi\rangle_2 = a|\psi_1\rangle_1 \otimes |\phi\rangle_2 + b|\psi_2\rangle_1 \otimes |\phi\rangle_2$$

$$|\psi\rangle_1 \otimes (a|\phi_1\rangle_2 + b|\phi_2\rangle_2) = a|\psi\rangle_1 \otimes |\phi_1\rangle_2 + b|\psi\rangle_1 \otimes |\phi_2\rangle_2$$

Let $\mathcal{H}_1 = \text{Span}\{|u_i\rangle_1 \mid i = 1, 2, \dots, N_1\}$, $\mathcal{H}_2 = \text{Span}\{|v_j\rangle_2 \mid j = 1, 2, \dots, N_2\}$, then $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$,

$$\dim \mathcal{H} = \dim \mathcal{H}_1 \cdot \dim \mathcal{H}_2 = N_1 N_2$$

$$\text{Let } |\psi\rangle_1 = \sum_i c_i |u_i\rangle_1, |\phi\rangle_2 = \sum_j d_j |v_j\rangle_2,$$

$$|\psi\rangle_1 \otimes |\phi\rangle_2 = \left(\sum_i c_i |u_i\rangle_1 \right) \otimes \left(\sum_j d_j |v_j\rangle_2 \right) = \sum_{i,j} c_i d_j |u_i\rangle_1 \otimes |v_j\rangle_2$$

Let $\mathcal{H}_1 = \text{Span}\{|u_1\rangle_1, |u_2\rangle_1\}$, $\mathcal{H}_2 = \text{Span}\{|v_1\rangle_2, |v_2\rangle_2\}$, then

$$|\psi\rangle_1 \otimes |\phi\rangle_2 = c_1 d_1 |u_1\rangle_1 \otimes |v_1\rangle_2 + c_1 d_2 |u_1\rangle_1 \otimes |v_2\rangle_2 + c_2 d_1 |u_2\rangle_1 \otimes |v_1\rangle_2 + c_2 d_2 |u_2\rangle_1 \otimes |v_2\rangle_2$$

For a state vector $|\chi\rangle = \frac{1}{\sqrt{2}}(|u_1\rangle_1 \otimes |v_1\rangle_2 + |u_2\rangle_1 \otimes |v_2\rangle_2) \in \mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$. Ask: do we have proper values of c_1, d_1, c_2, d_2 to match the requirement of $|\chi\rangle$? Answer: NO! $|\chi\rangle \neq |\psi\rangle_1 \otimes |\phi\rangle_2$.

$|\psi\rangle_1 \otimes |\phi\rangle_2$ is called tensor product state. $|\chi\rangle$ is called entangled state.

6.3.2 Entangled State

Eg.1 Kinds of Entangled State

$N = 2$, $\frac{1}{\sqrt{2}}(|u_1\rangle_1 \otimes |v_1\rangle_2 \pm |u_2\rangle_1 \otimes |v_2\rangle_2)$ is called Bell-state (Bell basis), or EPR state, or Schrödinger's cat state.

$N \geq 3$:

1. Werner state (W-state): robust
2. GHZ state: fragile
3. NOON state

Two spin system $(\frac{1}{2} \otimes \frac{1}{2})$. singlet: 1-entangled state. triplet: 2-tensor product state + 1-entangled state.

6.3.3 Inner product of tensor product

$$({}_1\langle\psi_2| \otimes {}_2\langle\phi_2|) (|\psi_1\rangle_1 \otimes |\phi_1\rangle_2) = ({}_1\langle\psi_2|\psi_1\rangle_1) ({}_2\langle\phi_2|\phi_1\rangle_2)$$

$$({}_1\langle u_i| \otimes {}_2\langle v_j|) (|u_k\rangle_1 \otimes |v_l\rangle_2) = ({}_1\langle u_i|u_k\rangle_1) ({}_2\langle v_j|v_l\rangle_2) = \delta_{ik}\delta_{jl}$$

If $\hat{A}$ acts only on $\mathcal{H}_1$, i.e. $\hat{A} \rightarrow |\psi\rangle_1$, and $\hat{B}$ acts only on $\mathcal{H}_2$, i.e. $\hat{B} \rightarrow |\phi\rangle_2$. Then $\hat{A} = \hat{A}_1 \otimes \hat{I}_2$, $\hat{B} = \hat{I}_1 \otimes \hat{B}_2$.

$$\hat{A}(|\psi\rangle_1 \otimes |\phi\rangle_2) = (\hat{A}_1 \otimes \hat{I}_2)(|\psi\rangle_1 \otimes |\phi\rangle_2) = \hat{A}_1 |\psi\rangle_1 \otimes \hat{I}_2 |\phi\rangle_2 = (\hat{A}_1 |\psi\rangle_1) \otimes |\phi\rangle_2$$

$$\hat{B}(|\psi\rangle_1 \otimes |\phi\rangle_2) = (\hat{I}_1 \otimes \hat{B}_2)(|\psi\rangle_1 \otimes |\phi\rangle_2) = \hat{I}_1 |\psi\rangle_1 \otimes \hat{B}_2 |\phi\rangle_2 = |\psi\rangle_1 \otimes (\hat{B}_2 |\phi\rangle_2)$$

The operator has no effect on the space on which it does not act.

Eg.2 Momentum

For $\hat{\mathbf{p}}^2 |\mathbf{p}\rangle = \mathbf{p}^2 |\mathbf{p}\rangle$, $|\mathbf{p}\rangle = |p_x\rangle \otimes |p_y\rangle \otimes |p_z\rangle$, $\hat{p}_x^2 = \hat{p}_x^2 \otimes \hat{I}_y \otimes \hat{I}_z$, the same as y, z .

Eg.3 Angular Momentum

Let $\hat{\mathbf{L}}^2 \rightarrow l$, $\hat{L}_z \rightarrow m_l$, $\hat{\mathbf{S}}^2 \rightarrow s$, $\hat{S}_z \rightarrow m_s$. $\mathcal{H}_l = |l, m_l\rangle$, $\mathcal{H}_s = |s, m_s\rangle$, $\mathcal{H} = \mathcal{H}_l \otimes \mathcal{H}_s$. Fix $l, s = 1/2$, then

$$\dim \mathcal{H} = \dim \mathcal{H}_l \cdot \dim \mathcal{H}_s = (2l + 1)(2s + 1) = 2(2l + 1)$$

Eg.4 Hydrogen Atom*

nL_j . n is principle, L_j is spectrum

6.4 Compatible Observable

1. Compatible observable (CO) is commuting observable, i.e. $[\hat{A}, \hat{B}] = 0$.
2. Common eigenstate (CE) or simultaneous eigenstate (SE)
3. Set of common eigenstate (SCE)
4. Complete set of commuting observable (CSCO)
5. Good quantum numbers (GQN)
6. Diagonal matrix. Block diagonal matrix. Diagonalization of matrix.

Thm of Compatible Observable:

$$[\hat{A}, \hat{B}] = 0 \iff \begin{cases} \hat{A}, \hat{B} \text{ have SCE} \\ \text{orthogonal basis} \end{cases}$$

$$[\hat{A}_i, \hat{A}_j] = 0, \quad (i, j = 1, 2, \dots, n) \iff \begin{cases} \textcircled{1} \text{ SCE (SSE),} \\ \textcircled{2} \text{ ON.} \end{cases}$$

$$[\hat{A}_i, \hat{A}_j] = 0, \quad [\hat{A}_i, \hat{B}] = 0, \quad (i, j = 1, 2, \dots, n) \implies \text{CSCO.}$$

diagonal matrix:

$$\hat{A} = \text{diag} \left(\underbrace{\lambda_1, \dots, \lambda_1}_{g_1}, \dots, \underbrace{\lambda_n, \dots, \lambda_n}_{g_n} \right), \quad g_i \ (i = 1, 2, \dots, n) \text{ degeneracy.}$$

block diagonal matrix:

$$\hat{B} = \text{diag} (B_1, B_2, \dots, B_n), \quad B_i : g_i \times g_i.$$

$$\begin{cases} \hat{L}^2 = \hbar^2 \text{diag} (0, 2, 2, 2, 6, 6, 6, 6, 6, \dots) . \\ \hat{L}_z = \hbar \text{diag} (0, -1, 0, 1, -2, -1, 0, +1, +2, \dots) . \end{cases}$$

[Proof]

Suppose

$$[\hat{A}, \hat{B}] = 0, \quad \hat{A}|\psi\rangle = \lambda|\psi\rangle.$$

Then

$$\hat{A}(\hat{B}|\psi\rangle) = \hat{B}\hat{A}|\psi\rangle = \lambda(\hat{B}|\psi\rangle).$$

Thus, $\hat{B}|\psi\rangle$ is eigenstate of $\hat{A}$ with the eigenvalue of λ .

① Suppose λ is non-degenerate. Then

$$\hat{B}|\psi\rangle = \mu|\psi\rangle \sim |\psi\rangle.$$

② If λ_n is degenerate, then

$$\hat{A}|\psi_n^i\rangle = \lambda_n|\psi_n^i\rangle, \quad i = 1, 2, \dots, g_n.$$

$$\langle\psi_n^i|\psi_m^j\rangle = \delta_{nm}\delta_{ij} \quad (n = 1, 2, \dots, g_n; j = 1, 2, \dots, g_m)$$

$$\begin{cases} \langle\psi_n^i|\hat{A}\hat{B}|\psi_m^j\rangle = \lambda_m\langle\psi_n^i|\hat{B}|\psi_m^j\rangle, \\ \langle\psi_n^i|\hat{B}\hat{A}|\psi_m^j\rangle = \lambda_n\langle\psi_n^i|\hat{B}|\psi_m^j\rangle \end{cases} \quad [\hat{A}, \hat{B}] = 0$$

$$(\lambda_n - \lambda_m)\langle\psi_n^i|\hat{B}|\psi_m^j\rangle = 0$$

$$n \neq m \implies \lambda_n \neq \lambda_m \implies \langle\psi_n^i|\hat{B}|\psi_m^j\rangle = 0 \implies \hat{B} \text{ is a block diagonal}$$

$$B_{ij}^{(n)} = \langle\psi_n^i|\hat{B}|\psi_n^j\rangle$$

$$\hat{B}^{(n)}|\varphi_n^i\rangle = \mu_i^{(n)}|\varphi_n^i\rangle \implies \langle\varphi_n^i|\hat{B}^{(n)}|\varphi_n^j\rangle = \mu_i^{(n)}\delta_{ij}$$

$\hat{B}^{(n)}$: $g_n \times g_n$ diagonal matrix.

$$|\varphi_n^i\rangle = \mathbb{I}|\varphi_n^i\rangle = \left(\sum_j |\psi_n^j\rangle\langle\psi_n^j|\right)|\varphi_n^i\rangle = \sum_j \left(\langle\psi_n^j|\varphi_n^i\rangle\right)|\psi_n^j\rangle$$

$$\implies \hat{A}|\varphi_n^i\rangle = \hat{A}\left[\sum_j \left(\langle\psi_n^j|\varphi_n^i\rangle\right)|\psi_n^j\rangle\right] = \lambda_n\left[\sum_j \left(\langle\psi_n^j|\varphi_n^i\rangle\right)|\psi_n^j\rangle\right] = \lambda_n|\varphi_n^i\rangle.$$

SCE of $\hat{A}$ and $\hat{B}$ $|\phi_{n,k}^i\rangle$

$$\begin{cases} \hat{A}|\varphi_{n,k}^i\rangle = \lambda_n|\varphi_{n,k}^i\rangle \\ \hat{B}|\varphi_{n,k}^i\rangle = \mu_k|\varphi_{n,k}^i\rangle \end{cases} \implies \begin{cases} \hat{A}\hat{B}|\varphi_{n,k}^i\rangle = \mu_k\lambda_n|\varphi_{n,k}^i\rangle \\ \hat{B}\hat{A}|\varphi_{n,k}^i\rangle = \lambda_n\mu_k|\varphi_{n,k}^i\rangle \end{cases} \implies [\hat{A}, \hat{B}]|\varphi_{n,k}^i\rangle = 0.$$

$$\forall |\psi\rangle,$$

$$|\psi\rangle = \sum_n C_n |\varphi_{n,k}^i\rangle \Rightarrow [\hat{A}, \hat{B}]|\psi\rangle = \sum_n C_n [\hat{A}, \hat{B}]|\varphi_{n,k}^i\rangle = 0 \Rightarrow [\hat{A}, \hat{B}] = 0.$$

6.5 Representation Transformation

6.5.1 Wavefunction

The wave-function is the projection of the state-vector in a certain rep.

$$\hat{\mathbf{x}}\text{-rep: } \hat{\mathbf{x}}|\mathbf{x}\rangle = \mathbf{x}|\mathbf{x}\rangle. \begin{cases} \langle\mathbf{x}|\mathbf{x}'\rangle = \delta(\mathbf{x} - \mathbf{x}') & \text{ON} \\ \int d^3x |\mathbf{x}\rangle \langle\mathbf{x}| = \hat{I} & \text{RI} \end{cases}$$

$$|\psi\rangle = \hat{I}|\psi\rangle = \left(\int d^3x |\mathbf{x}\rangle \langle\mathbf{x}| \right) |\psi\rangle = \int d^3x |\mathbf{x}\rangle \psi(\mathbf{x})$$

where $\psi(\mathbf{x}) = \langle\mathbf{x}|\psi\rangle$ is wavefunction in $\mathbf{x}$ -rep.

$$\hat{\mathbf{p}}\text{-rep: } \hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle. \begin{cases} \langle\mathbf{p}|\mathbf{p}'\rangle = \delta(\mathbf{p} - \mathbf{p}') & \text{ON} \\ \int d^3p |\mathbf{p}\rangle \langle\mathbf{p}| = \hat{I} & \text{RI} \end{cases}$$

$$|\psi\rangle = \left(\int d^3p |\mathbf{p}\rangle \langle\mathbf{p}| \right) |\psi\rangle = \int d^3p |\mathbf{p}\rangle \tilde{\psi}(\mathbf{p})$$

where $\tilde{\psi}(\mathbf{p}) = \langle\mathbf{p}|\psi\rangle$.

$$|\psi\rangle = \left(\sum_j |u_n\rangle \langle u_n| \right) |\psi\rangle = \sum_n c_n |u_n\rangle$$

where $c_n = \langle u_n|\psi\rangle$ is the wavefunction in $\{|u_n\rangle\}$ -rep or $\hat{A}$ -rep, if $\hat{A}|u_n\rangle = A_n|u_n\rangle$.

6.5.2 Representation Transformation

$$|\psi\rangle = \int d^3p |\mathbf{p}\rangle \tilde{\psi}(\mathbf{p}) = \int d^3x \underbrace{\left(\int d^3p |\mathbf{p}\rangle \langle\mathbf{p}| \right)}_{=\hat{I}} \psi(\mathbf{x}) = \int d^3p |\mathbf{p}\rangle \int d^3x \psi(\mathbf{x}) \langle\mathbf{p}|\mathbf{x}\rangle$$

$$\Rightarrow \tilde{\psi}(\mathbf{p}) = \int d^3x \psi(\mathbf{x}) = \psi_{\mathbf{p}}^*(\mathbf{x})$$

where $\psi_{\mathbf{p}}(\mathbf{x}) = \langle\mathbf{x}|\mathbf{p}\rangle$.

$$\psi(\mathbf{x}) = \tilde{\psi}(\mathbf{p})\psi_{\mathbf{p}}(\mathbf{x})$$

The key is $\langle\mathbf{x}|\mathbf{p}\rangle$ and $\langle\mathbf{p}|\mathbf{x}\rangle$

In $\hat{\mathbf{x}}$ -rep,

$$\langle \mathbf{x} | \mathbf{p} \rangle = \psi_{\mathbf{p}}(\mathbf{x}) = N e^{i\mathbf{p} \cdot \mathbf{x} / \hbar} \quad (\Longleftrightarrow \hat{\mathbf{p}} = -i\hbar \nabla)$$

where N is the normalization factor.

In $\hat{\mathbf{p}}$ -rep, $\hat{\mathbf{x}} = +i\hbar \nabla_{\mathbf{p}} \Longleftrightarrow \langle \mathbf{p} | \mathbf{x} \rangle = N e^{-i\mathbf{p} \cdot \mathbf{x} / \hbar}$.

How to determine the normalization factor?

$$\begin{aligned} \langle \mathbf{x} | \mathbf{x}' \rangle &= \delta(\mathbf{x} - \mathbf{x}') = \int d^3p \langle \mathbf{x} | \mathbf{p} \rangle \langle \mathbf{p} | \mathbf{x}' \rangle = N^2 \int d^3p e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}') / \hbar} \\ \int_{-\infty}^{\infty} e^{ikx} dk &= 2\pi \delta(x) \implies \langle \mathbf{x} | \mathbf{x}' \rangle = M^2 (2\pi\hbar)^3 \delta(\mathbf{x} - \mathbf{x}') \implies N = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} \\ \implies \langle \mathbf{x} | \mathbf{p} \rangle &= \psi_{\mathbf{p}}(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{i\mathbf{p} \cdot \mathbf{x} / \hbar} \quad (\Longleftrightarrow \hat{\mathbf{p}} = -i\hbar \nabla) \end{aligned}$$

$\hat{\mathbf{x}}$ -representation	$\hat{\mathbf{p}}$ -representation
$\hat{\mathbf{x}} = \hat{\mathbf{x}}, \quad \hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{x}}$	$\hat{\mathbf{p}} = \hat{\mathbf{p}}, \quad \hat{\mathbf{x}} = +i\hbar \nabla_{\mathbf{p}}$
$\hat{\mathbf{x}} \mathbf{x} \rangle = \mathbf{x} \mathbf{x} \rangle$	$\hat{\mathbf{p}} \mathbf{p} \rangle = \mathbf{p} \mathbf{p} \rangle$
$\hat{\mathbf{x}} \delta(\mathbf{x} - \mathbf{x}') = \mathbf{x}' \delta(\mathbf{x} - \mathbf{x}')$	$\hat{\mathbf{p}} \delta(\mathbf{p} - \mathbf{p}') = \mathbf{p}' \delta(\mathbf{p} - \mathbf{p}')$
$\psi_{\mathbf{p}}(\mathbf{x}) = \langle \mathbf{x} \mathbf{p} \rangle = \left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{i\mathbf{p} \cdot \mathbf{x} / \hbar}$	$\psi_{\mathbf{x}}(\mathbf{p}) = \langle \mathbf{p} \mathbf{x} \rangle = \left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{-i\mathbf{p} \cdot \mathbf{x} / \hbar}$

6.5.3 Schrödinger eq

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle = \left[\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right] |\psi(t)\rangle$$

In $\hat{\mathbf{x}}$ -rep,

$$\implies i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}) \psi(\mathbf{x}, t)$$

where $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$.

In $\hat{\mathbf{p}}$ -rep,

$$\implies i\hbar \frac{\partial}{\partial t} \tilde{\psi}(\mathbf{p}, t) = \frac{\hat{\mathbf{p}}^2}{2m} \tilde{\psi}(\mathbf{p}, t) + V(i\hbar \nabla_{\mathbf{p}}) \tilde{\psi}(\mathbf{p}, t)$$

where $\tilde{\psi}(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle$.

6.6 Adjoint

$\hat{A}^\dagger = (\hat{A}^T)^*$ 不一定对每个算符成立, 有些算符不能写成矩阵. 原始定义为

$$(\hat{A} |\psi\rangle)^\dagger = \langle \psi | \hat{A}^\dagger \Longleftrightarrow \langle \psi | \hat{A}^\dagger | \phi \rangle = \langle \phi | \hat{A} | \psi \rangle^*$$

Pf $|\psi\rangle = \sum_n c_n |u_n\rangle, \quad |\phi\rangle = \sum_m d_m |u_m\rangle.$

$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \sum_{n,m} \langle u_n|c_n^*\hat{A}^\dagger d_m|u_m\rangle = \sum_{n,m} c_n^* d_m \langle u_n|\hat{A}^\dagger|u_m\rangle = \sum_{n,m} c_n^* d_m (\hat{A}^\dagger)_{nm}.$$

$$\begin{aligned} (\langle\phi|\hat{A}|\psi\rangle)^* &= \left[\sum_{n,m} \langle u_m|d_m^*\hat{A}c_n|u_n\rangle \right]^* = \left[\sum_{n,m} d_m^* c_n (\hat{A})_{mn} \right]^* = \sum_{n,m} d_m c_n^* (\hat{A})_{mn}^* \\ \implies (\hat{A}^\dagger)_{nm} &= (\hat{A})_{mn}^* = (\hat{A}^\top)_{nm}^*. \implies \hat{A}^\dagger = (\hat{A}^\top)^*. \end{aligned}$$

Adjoint = transpose + complex conjugate.

6.7 Unitary and Anti-unitary

$$\text{space (spacial)} \begin{cases} \text{translation} & \text{momentum conservation} \\ \text{reflection (parity inversion)} & \text{party conservation} \\ \text{rotation} & \text{angular momentum conservation} \end{cases}$$

$$\text{time (temporary)} \begin{cases} \text{translation} & \text{energy} \\ \text{reversal} & \text{no conservation but Kramer's degeneracy} \end{cases}$$

Time reversal is anti-linear and anti-unitary.